MILITARY SPECIFICATION

INDICATOR, ANALOG DISPLAY

IP-722F/AVA-1

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, performance, and acceptance requirements for the Analog Display Indicator (ADI), IP-722F/AVA-1. The indicator generates and displays a contact analog presentation which includes a flight command presentation. The indicator also accepts and displays Radar Data Converter (RDC) and Forward Looking Infrared (FLIR) information.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.2 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-1 Electron Tube, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addessed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ. 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5895

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MIL-C-675	Coating of Glass Optical Elements (Anti-Reflection)
MIL-B-5087	Bonding, Electrical and Lightning Protection for Aerospace Systems
MIL-W-5088	Wiring; Aerospace Vehicle
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronics, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-T-81624	Television Display Requirements for Amendment 1 the Weapons System for Guided Weapon Mark 1 All Mods (WALLEYE) and Guided Weapon, Practice MK 4 MOD 0/1

STANDARDS

MILITARY

MIL-STD-461	Electromagnetic Emissions and Susceptability Requirements for the Control of Electromagnetic Interference

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MIL-STD-470	Maintainability Program for System and Equipment
MIL-STD-471	Maintainability Verification/ Demonstration/Evaluation
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Design and Production Acceptance Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-882	System Safety Program Requirements
MIL-STD-965	Parts Control Program
MIL-STD-2074	Failure Classification for Reliability Testing
MS 17322	Meter, Time Totalizing, Miniature Digital 115V, 400 Cycle

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified, (see 6.2.1), a sample(s) shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Parts, materials, and processes. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall be in accordance with the requirements specified herein.

b. Other parts and materials requirements shall be in accordance with MIL-E-5400.

c. Nonrepairable subassemblies shall be used in accordance with MIL-E-5400.

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d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard parts and materials approval. Approval for the use of non-standard parts and materials (including electron tubes, transistors, and diodes) other than microelectronic devices shall be in accordance with MIL-E-5400. Approval for microelectronic devices shall be in accordance with MIL-STD-965.

3.2.2 Limited-life items. The ADI shall not contain Limited Life Items which require scheduled replacement within the minimum specified operating life of the ADI.

3.2.3 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall meet the requirements of MIL-STD-965.

3.2.4 Modules. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly. Modules shall not be hardwired.

3.3 General design and construction. Unless otherwise specified herein, the ADI shall be in accordance with all the applicable design, construction, and workmanship requirements of MIL-E-5400.

3.3.1 Total weight. The total weight of the ADI, including light filters but excluding cables, shall be a minimum consistent with good design and shall be not greater than 47 pounds (21.3 kg). The total weight shall include provisions for mounting.

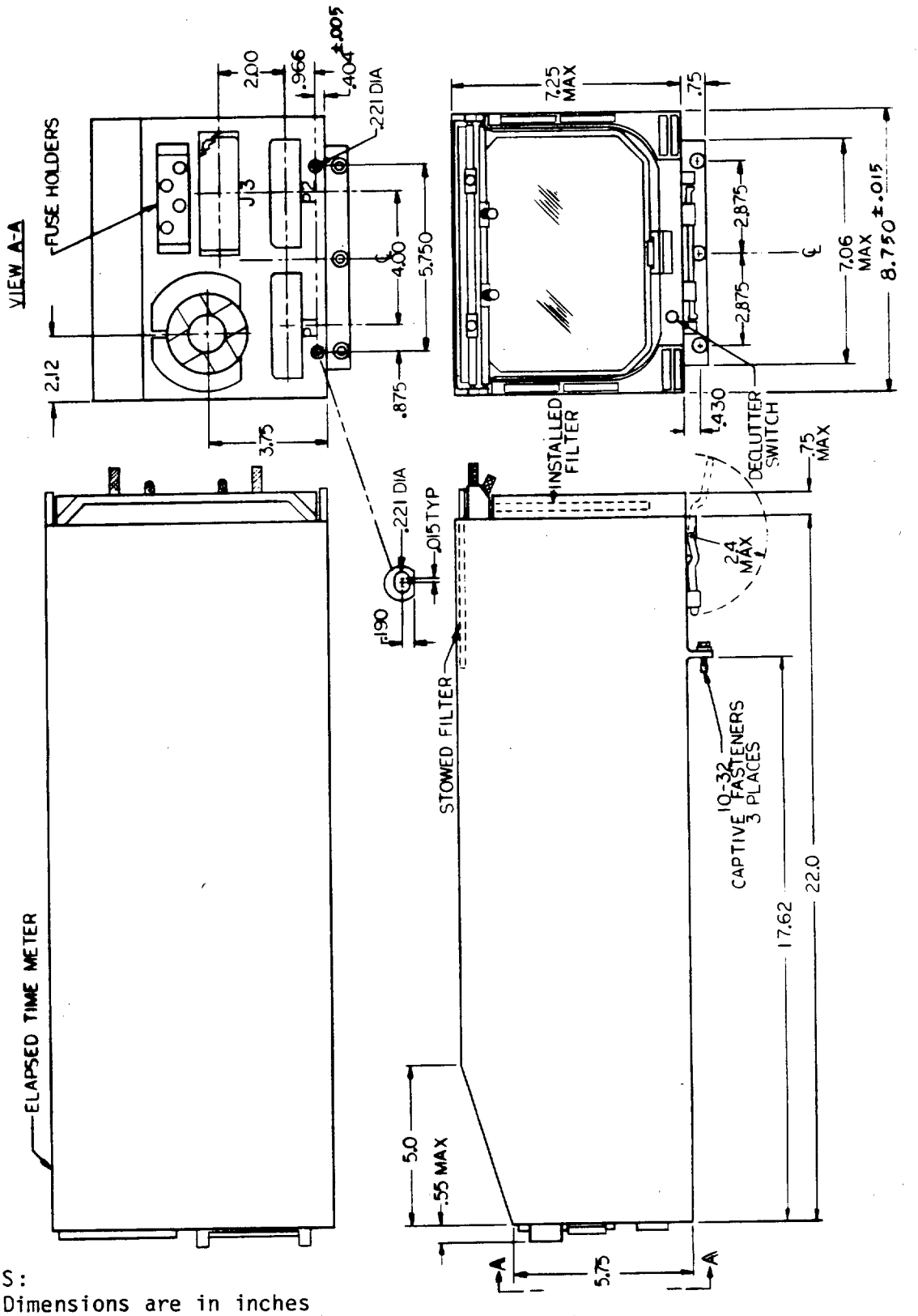
3.3.1.1 Size. The dimensions of the ADI shall be in accordance with figures 1, 2, and 3.

3.3.1.2 Mounting. Mounting provisions for the ADI shall be in accordance with figure 1.

3.3.1.3 Time totalizing meter. The ADI shall contain a time totalizing meter, MS 17322, in accordance with MIL-M-7793. The meter shall be mounted as shown in figure 1.

3.3.2 Reliability program. The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a reorder from a contractor who is producing the ADI, the program previously approved may be continued unless otherwise specified in the contract or order.

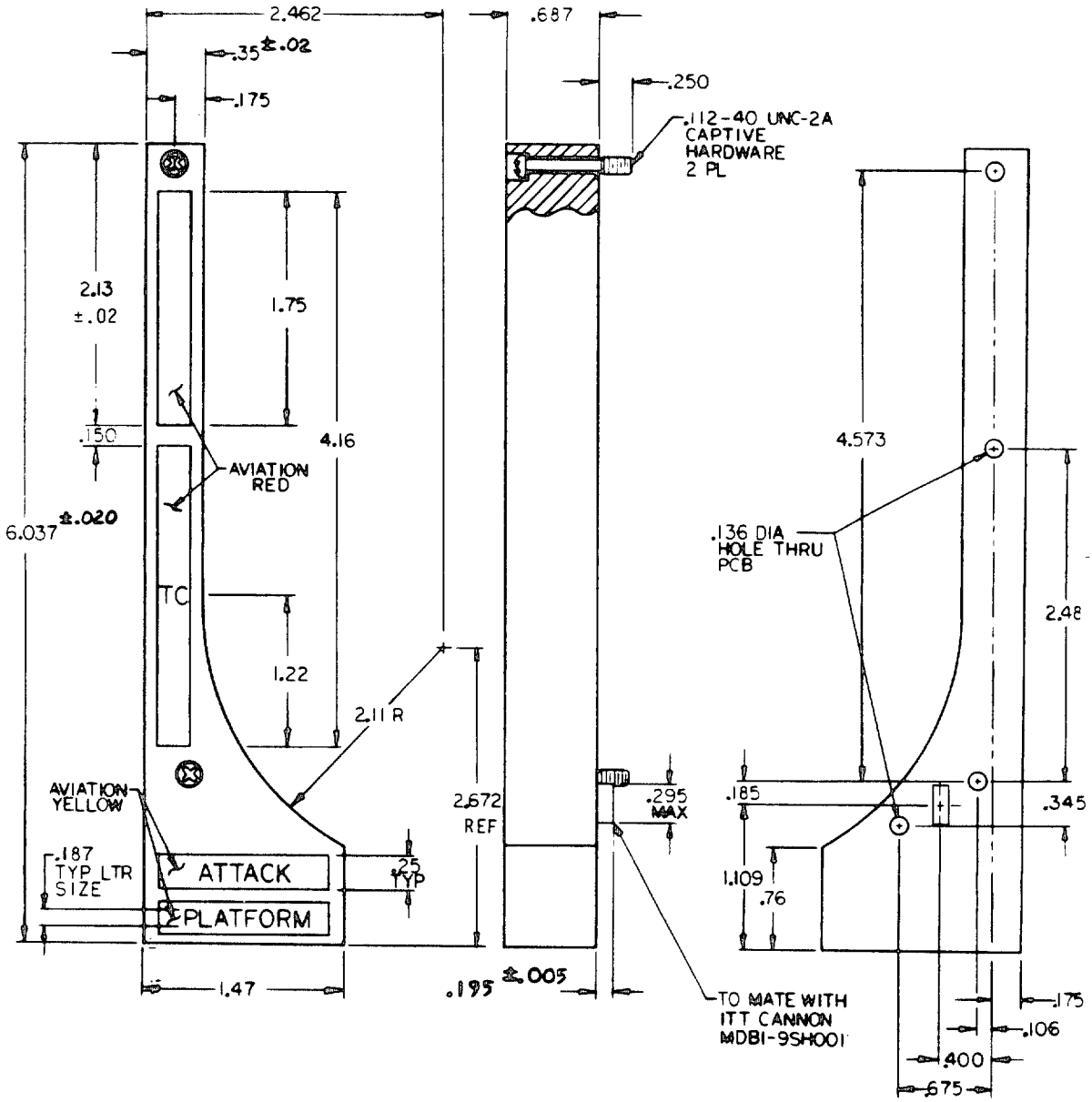
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NOTES:
1. Dimensions are in inches
2. Unless otherwise noted, tolerances are ±0.010.

FIGURE 1. Analog display indicator.

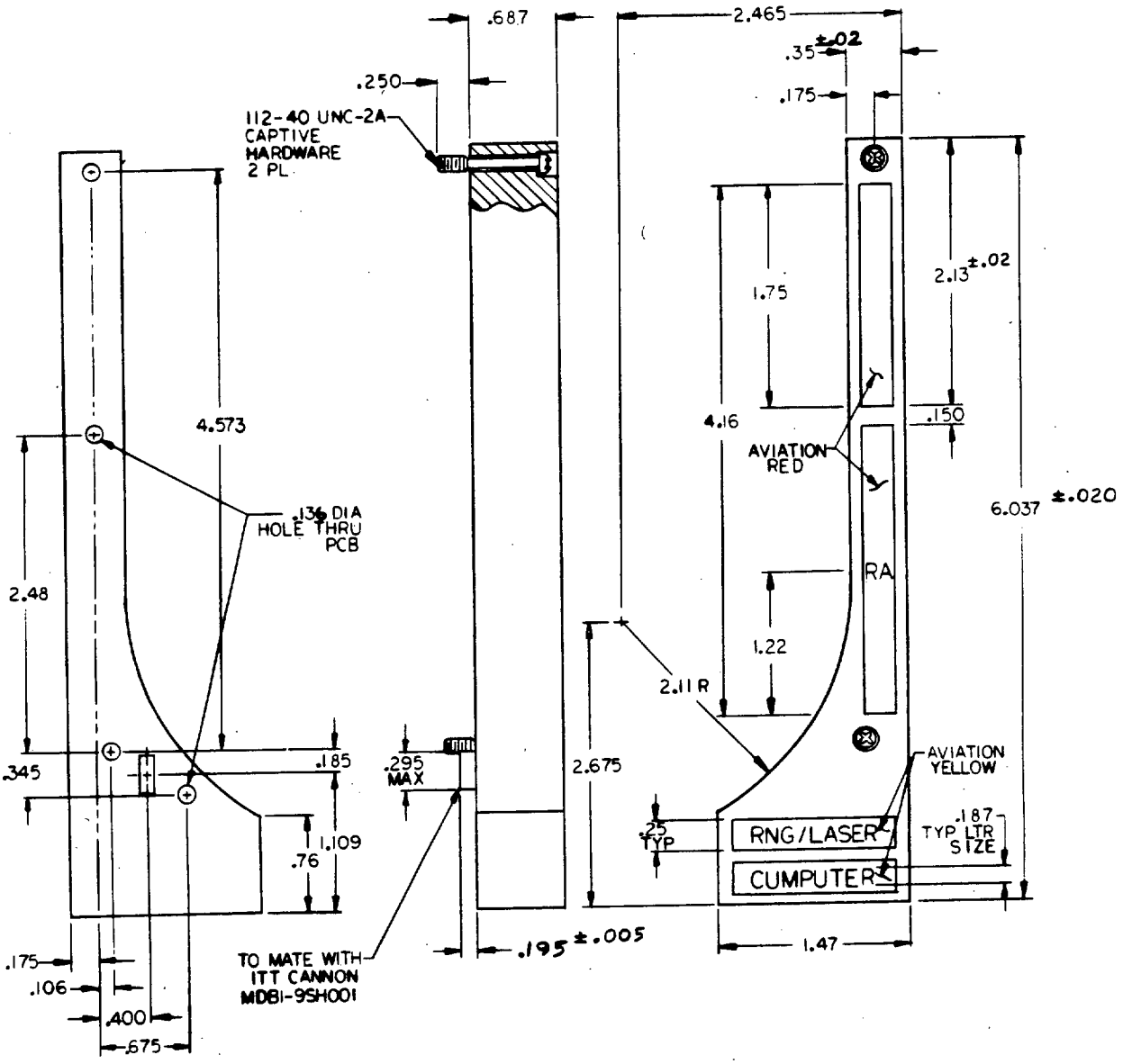
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Notes:
1. Dimensions are in inches.
2. Unless otherwise noted, tolerances are ± 0.010.

FIGURE 2. Left indicator lights.

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Notes:
1. Dimensions are in inches.
2. Unless otherwise noted, tolerances are ± 0.010.

FIGURE 3. Right indicator lights.

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3.3.2.1 Operational stability. The ADI shall operate with satisfactory performance, continuously or intermittently without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The ADI shall have a total operating life of 10,000 hours with minimal servicing and replacement of parts.

3.3.2.3 Reliability. The ADI, including any Built-In-Test (BIT) provisions, shall have an upper test Mean Time Between Failure (MTBF) ($\emptyset_0$) of 445 hours with 80 percent confidence when tested and accepted as specified in 4.5.1.3.

3.3.2.4 Storage service life. At any time within, or immediately following an 18 month storage period, the ADI shall be capable of meeting all the requirements of this specification, under any combination of environments specified herein without component or part replacement, readjustment, or other maintenance actions.

3.3.3 Cabling and connections.

3.3.3.1 Cables and connectors. The ADI shall provide for the use of wiring and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection cabling. The ADI shall be capable of specified operation using external wiring in accordance with the applicable requirements of MIL-W-5088. Such external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements.

3.3.4 Lighting and controls.

3.3.4.1 Controls. ADI controls, control knobs, and legends shall be in accordance with MIL-E-5400.

3.3.4.2 Indicator lights. Unless otherwise specified herein, the ADI lights shown in figures 2 and 3 shall be in accordance with MIL-E-5400.

3.3.5 Interchangeability. The ADI shall be in accordance with the interchangeability requirements of MIL-E-5400.

3.3.6 Interference control. The generation of electromagnetic interference (EMI) by the ADI and the vulnerability of the ADI to EMI shall be in accordance with the requirements of MIL-STD-461.

3.3.6.1 Conducted and radiated emissions. The conducted and radiated emissions of the ADI shall be controlled in accordance with the limits and requirements of MIL-STD-461 and as specified herein.

3.3.6.2 Conducted and radiated susceptibility. There shall be no malfunction, change in normal indication or degradation of performance of the ADI when subjected to the tests and requirements specified in MIL-STD-461.

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3.3.7 Maintainability. The maintainability program, BIT features, construction and packaging, provisions for test points, and other maintainability parameters shall be in accordance with MIL-STD-470 and as specified herein.

3.3.7.1 Detail maintainability requirements.

3.3.7.1.1 Preventive maintenance. The ADI shall not have preventive maintenance requirements.

3.3.7.1.2 Organizational level. Organizational level maintenance shall use test mode (see 3.4.13.4.5) (BIT) for fault isolation to the aircraft weapon replaceable assembly (WRA). This level of maintenance shall remove the defective ADI, replace it with a ready for issue ADI and verify system repair using the test mode. Ground support equipment (GSE) shall not be required.

3.3.7.1.2.1 Fault isolation. BIT shall be incorporated into the ADI to enable the unambigous fault isolation to the defective aircraft WRA by means of test mode analysis for at least 95 percent of all possible ADI faults, when the faults are ranked by frequency of occurrence, with a 90 percent confidence level.

3.3.7.1.2.2 Remove and replace. The time to remove and replace the equipment in the aircraft shall be not greater than 12 minutes mean time (exclusive of access time).

3.3.7.1.2.3 Turn around time. The organizational level turn around time shall be the total of the time required for fault isolation, removal and replacement, and repair verification. The turn around time shall be not greater than 16 minutes mean time.

3.3.7.1.3 Intermediate level. Intermediate level maintenance shall consist of the repair of the ADI and modules, shop replaceable assemblies (SRA), in off-aircraft shops.

3.3.7.1.3.1 ADI repair. The ADI shall be compatible with Mini-Shop Automatic Checkout Equipment (Mini-SACE AN/ASM-461) for fault detection, isolation to the SRA, removal and replacement of the SRA, and ADI operational verification.

3.3.7.1.3.2 SRA repair. SRAs shall have test points available which allow fault detection and isolation to the faulty piece part group. Repair is achieved by the removal and replacement of the identified piece part(s). SRA operational verification is accomplished by test point examination and by the operation of the SRA in the ADI.

3.3.7.1.3.3 Intermediate level equipment repair time. The intermediate level Equipment Repair Time (ERT) shall be the total of the times for removal and replacement of the defective SRA and the removal and replacement of a defective part(s) including fault isolation, access, and secure times. The ERT shall be not greater than 110 minutes.

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3.3.7.2 General maintainability requirements.

3.3.7.2.1 Task criteria. Each of the maintenance times specified in 3.3.7.1.2 shall be accomplished with a minimum of two maintenance personnel of no higher than intermediate skill (E-5) for at least 95 percent of the maintenance tasks.

3.3.7.2.2 Built-In-Test. Suitable BIT features shall be incorporated into the ADI to provide the operator with indications of ADI readiness and to locate the fault to the malfunctioning aircraft WRA. These features shall be implemented using manually activated test pattern(s) displayed on the cathode ray tube (CRT) and shall be capable of unambiguously isolating to the defective aircraft WRA for at least 95 percent of all possible ADI faults when the faults are ranked in order of their occurrence with a 90 percent confidence level.

3.3.7.2.2.1 Test display. During the test mode (see 3.4.13.4.5), the ADI shall provide a static or dynamic test display. The test display shall exercise the complete capabilities of the ADI and unambiguously isolate to the WRA. The test display shall be easily interpreted by the operator and shall not require complex mental calculations.

3.3.7.2.3 Test points. The ADI shall have test points.

3.3.7.2.3.1 Functional test point arrangement. Test points shall be grouped within a multipin connector which is readily accessible at the level of maintenance for which it is intended. The test points shall be grouped by the functional maintenance level requirements as follows:

a. WRA test points shall be located on exterior surfaces of the ADI so as to be readily accessible for maintenance.

b. SRA test points shall be located in such internal positions so as to be readily accessible for maintenance.

3.3.7.2.4 Shop replaceable assembly orientation. SRAs shall be oriented in the ADI such that when the ADI is mounted in the aircraft, intermittent operation due to catapult launches and arrested recoveries is precluded.

3.3.7.2.5 Extractors. SRAs shall have attached extractors which utilize mechanical leverage in order to simplify SRA insertion, withdrawal and replacement. It shall not be necessary to restrain the ADI chassis in order to remove or install SRAs. The extractors shall also provide a positive locking device for the SRAs to assure proper seating of the connector pins and to restrain the SRA in the ADI. SRAs shall not be restrained by resilient pads on access covers.

3.3.8 Nomenclature, nameplates and identification marking. Nomenclature, serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.3.9 Standard conditions. The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the ADI.

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a. Temperature	Room ambient (25 ± 10°C)
b. Altitude	Normal ground
c. Vibration	None
d. Humidity	Room ambient to 90 percent relative humidity
e. Voltage	115 ± 1.0 VAC, 3 Phase, 400 ± 20 Hz and 27.5 ± 0.5 VDC

3.3.10 Warm-up time. The time required for the ADI to warm up prior to operation shall be kept to a minimum and shall be not greater than two minutes under standard conditions and three minutes at extreme service conditions.

3.3.11 Service conditions. Except as modified herein, the ADI shall operate as specified under any of the environmental service conditions or reasonable combination of these conditions in accordance with MIL-E-5400 for Class 1 equipment.

3.3.11.1 Vibration. The ADI shall operate as specified when subjected to the vibration requirements of MIL-E-5400, curve I of figure 2, except at a maximum vibration level of 5g shall apply.

3.3.11.2 Temperature. The ADI shall meet the temperature requirements for Class 1 equipment in accordance with MIL-E-5400, except at temperatures within the range +56°C to +71°C, operation without permanent failure is required rather than operation within the specification tolerances.

3.3.11.3 Altitude. The ADI shall meet the altitude requirements for Class 1 equipment in accordance with MIL-E-5400, except the maximum altitude shall be 55,000 feet.

3.3.11.4 Shock. The ADI shall be capable of meeting the performance requirements when subjected to shock inputs in accordance with MIL-E-5400. The shaped pulse of these shocks shall be half sine wave.

3.3.11.5 Acceleration. The ADI shall meet the performance requirements when subjected to acceleration inputs in accordance with procedure III of MIL-E-5272.

3.3.12 Service conditions - electrical.

3.3.12.1 Primary input power. The ADI shall be designed to meet all applicable requirements of MIL-E-5400 and shall give specified performance from the following power sources with characteristics and limits in accordance with MIL-STD-704. The power required shall be not greater than the specified amount.

a. AC power: (Three phase) 115/200 V, 400 Hz, 400 VA maximum.
b. DC power: 28 V, 50 watts maximum.

3.3.12.2 Lighting power. Input power for lighting shall be not greater than 1.0 ampere at +28 VDC.

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3.3.12.3 Overload protection. Overload protection for the ADI shall be provided in the ADI. All parts and circuits of the ADI which are likely to carry an overload due to any failure or poor adjustments shall have suitable overload protective devices or shall be derated to withstand such overload without permanent damage to the ADI. The use of fuses or other protective devices shall be held to a minimum.

3.3.12.4 Under voltage protection. The ADI shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3.12.5 Loss of power protection. A 400 Hz single phase or a balanced loading of Y connected three phase power shall be used. The ADI shall not suffer damage nor shall it cause heavy loading on the remaining phases, if one of the phase leads does not supply power to the ADI.

3.3.13 Surrounding air pressure. For design purposes, air pressure surrounding the ADI shall be assumed equivalent to free stream static pressure. The ADI shall withstand or operate at 1.33 psia surrounding air pressure.

3.3.13.1 Normal. The surrounding air pressure may vary at a rate as high as 0.25 psi per second between 21.0 and 2.73 psia.

3.3.13.2 Abnormal-explosive decompression.

3.3.13.2.1 Condition II. The ADI shall meet functional requirements (including accuracy requirements) immediately after being subjected to an explosive decompression in which the surrounding air pressure varies with time in accordance with the curve specified in Condition I on figure 27.

3.3.13.2.2 Condition II. The ADI shall not deform or disintegrate in a manner which could cause injury to a crew member or otherwise jeopardize aircraft safety when subjected to an explosive decompression in which the surrounding air pressure varies with time in accordance with the curve specified in Condition II on figure 27. The ADI shall not be required to function after experiencing this condition.

3.3.14 Thermal requirements and cooling. The ADI shall meet thermal design requirements in accordance with MIL-E-5400.

3.3.14.1 Thermal design. The ADI shall not require cooling by means other than conduction through the mounting hardware and convection into the cabin environment. In the event that air vents are required, they shall be protected as necessary to preclude the entry of foreign material. A fan, integral with the ADI, may be provided to maximize convection cooling.

3.3.14.2 Heat dissipation. The heat dissipation for the ADI shall be not greater than 365 watts, exclusive of cooling fan.

3.3.14.3 Exterior (touch) temperature.

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3.3.14.3.1 Temperature at standard conditions. Under standard conditions of operation specified herein, the ADI (touch) temperature shall be not greater than 60°C. The stabilized surface temperature of the front panel and display face shall be not greater than 43°C.

3.3.14.3.2 Temperature at operating conditions. While the ADI is being operated between sea level and 25,000 feet and at nominal air temperature of 25°C on the pilot's side of the display and 55°C nominal on the back side, the stabilized surface temperature shall be not greater than 63°C for the display face.

3.3.15 Human factors engineering. The requirements for applying the principles and criteria of human factors engineering shall be in accordance with MIL-E-5400.

3.3.16 Bonding. Bonding shall be in accordance with MIL-B-5087.

3.3.17 Safety. The ADI shall be designed to preclude the incorporation of features which result in critical or catastrophic hazards in accordance with MIL-STD-882. Personnel safety shall be in accordance with MIL-E-5400.

3.3.18 Coating of optical surfaces. Excepting the CRT and as otherwise specified herein, all optical surfaces shall be coated with nonreflective protective coatings if they are to transmit light and with reflective coatings if they are to reflect light. These coatings shall be in accordance with MIL-C-675.

3.3.18.1 Incident solar radiation. Suitable protection shall be incorporated in the ADI such that light energy resulting from steady-state solar illumination from any angle shall not damage nor degrade the CRT.

3.3.19 Fungus treatment. The ADI shall be fungus proofed in accordance with MIL-E-5400.

3.3.20 Moisture treatment. The ADI shall be treated for moisture in accordance with MIL-E-5400.

3.3.21 Acoustic noise generation. The noise level generated by the ADI shall be not greater than 85 decibels (reference to 0.0002 dynes per square centimeter) measured at any point two feet from the ADI.

3.3.22 Gray scale. The ADI shall be capable of displaying at least six shades of gray in order to display group 1 and group 2 video (see 3.4.13.4.3). There shall be a minimum separation, in brightness ratio, of 1.4 between shades in a 10,000 foot-lambert solar ambient condition. The following shades are applicable:

a. Black - Zero excitation or a negative excitation.

b. Medium dark - Two shades lighter than black, minimum.

c. Medium light - One shade lighter than medium dark, minimum.

d. Bright - Two shades lighter than medium light, minimum.

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3.4 Performance. Unless otherwise specified herein, values set forth to establish the requirements of specified performance apply to performance under both standard and extreme service and all input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions shall be specified.

3.4.1 Operation. The ADI shall provide the pilot with a dynamic display in Contact Analog (CA) mode, simulating the view seen through a forward cockpit window with a visibility of 60° in elevation and 50° in azimuth. To simulate actual contact flight, the display shall provide a flight path presentation on a field of ground and sky textures. This presentation in combination with other visual symbols made available by the ADI shall present information to the pilot in take-off, flying, attack, and landing modes independent of flight visibility conditions. The ADI shall also be capable of operation in Terrain Clearance (TC) mode which shall produce a presentation that is primarily dependent upon processed search radar data received from the RDC. In the TC mode, the display shall be simplified to provide the pilot with information for avoiding terrain obstacles during low-level flight. The ADI shall also be capable of accepting and displaying FLIR or external video while in the target recognition attack multisensor (TRAM) mode.

3.4.2 Search radar terrain clearance display. The ADI shall provide circuitry suitable for processing the RDC information so as to present a search radar TC display. The ADI shall video-mix all symbols listed in 3.4.13.4.3(a) and the offset impact point marker so that they appear superimposed on the search radar terrain video. The indicator shall also video-mix and superimpose the pitch trim cursor when input 33(b), table VII is present. When in the TC mode, the ADI shall switch to CA mode for aircraft roll angles between 110° and 250°.

3.4.3 Switching time. The time required to switch to (or from) a stable analog display of the CA mode from (or to) one of the RDC displays of the TC mode after a discrete switching signal (input 23) is received from the control panel, shall not exceed 200 milliseconds. The time required to switch from (or to) CA or TC to (or from) FLIR mode after a discrete switching signal is received shall not exceed 1.0 second. The display shall be blanked for mode transistions that require the yoke to roll. The blanking time for rotational mode transistions shall minimally equal roll stabilization time.

3.4.4 Angular coverage. The display, in CA mode, shall include a total of 60 degrees of pitch and 50 degrees of azimuth at zero degree roll.

3.4.5 Attitude range. The display shall be capable of a continuous true (within one degree) presentation of the airplane's change in attitude through 360 degress of azimuth and roll and ±90 degrees in pitch. There shall be no apparent display discontinuity during a loop maneuver.

3.4.5.1 Roll commutator switch. A roll commutator switch shall be provided in the ADI which shall be actuated when the airplane exceeds a roll angle of ±110 (±10) degrees from wings level fight. This switch contact shall be installed so as to allow utilization of the +28 VDC signal specified in 3.4.13.1.8. When in a command over-the-shoulder maneuver, the raster shall rotate 180° at 90° of elevation and the flight path shall invert when the roll commutator switch

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passes +110 (±10) degrees roll. When not in a command over-the-shoulder maneuver and at a 90° elevation angle, the roll +28 VDC signal specified in 3.4.13.1.8 is not present. The raster shall rotate 180° due to inertial navigation system (INS) gimbal flip at 90°. The path shall rotate with the raster indicating that this is not a command maneuver.

3.4.6 Inertial subsystem reliability indicator. An indicator shall be located at the lower left-hand corner of the analog display front panel as shown in figures 1 and 2. The indicator shall contain three lamps and shall be aviation yellow in color with a minimum light output in bright areas of 150 foot-lamberts. This indicator shall be marked PLATFORM and shall be illuminated by a +28 VDC signal on connector pin P2-50. See table VII, input 43. In addition, the indicator shall illuminate when there is a 2° or greater lag between the roll attitude input information and the roll servo. The indicator shall not be dimmed by the pilot's control panel dimmer switch.

3.4.7 Computer reliability indicator. An indicator shall be located at the lower right-hand corner of the analog display panel as shown in figures 1 and 3. The indicator shall contain three lamps and shall be aviation yellow in color with a minimum light output in bright areas of 150 foot-lamberts. The indicator shall be marked COMPUTER and shall be illuminated by a +28 VDC signal on connector pin P1-25. See table VII, input 47. The indicator shall be capable of being dimmed by the pilot's control panel dimmer switch.

3.4.8 Search radar terrain clearance reliabilty indicator. An indicator shall be located on the left side of the analog display panel as shown in figures 1 and 2. The indicator shall contain five lamps and shall be aviation red in color with a minimum light output in bright areas of 300 foot-lamberts. The indicator shall be marked TC and shall be illuminated by a +28 VDC signal on connector pin P1-49. See table VII, input 1. The indicator shall be capable of being dimmed by the pilot's control panel dimmer switch.

3.4.9 Altimeter reliability indicator. An indicator shall be located on the right side of the analog display panel as shown in figures 1 and 3. The indicator shall contain five lamps and shall be aviation red in color with a minimum light output in bright areas of 300 foot-lamberts. The indicator shall be marked RA and shall be illuminated by a +28 VDC signal on connector pin P1-58. See table VII, input 5. The indicator shall be capable of being dimmed by the pilot's control panel dimmer switch.

3.4.10 RNG/LASER indicator. An indicator shall be located above the COMPUTER indicator as shown in figure 3. The indicator shall contain three lamps and shall be aviation yellow in color with a minimum light output in bright areas of 150 foot-lamberts. The indicator shall be marked RNG/LASER and shall be illuminated by a +28 VDC signal on connector pin P1-93. See table VII, input 8. The indicator shall be capable of being dimmed by the pilot's control panel dimmer switch.

3.4.11 Attack indicator. An indicator shall be located above the PLATFORM indicator as shown in figure 2. The indicator shall contain three lamps and shall be aviation yellow in color with a minimum light output in bright areas of 150 foot-lamberts. The indicator shall be marked ATTACK and shall be illuminated by a +28 VDC signal on connector pin P1-92. See table VII, input 7. The indicator shall be capable of being dimmed by the pilot's control panel dimmer switch.

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3.4.12 Breakaway indicators. There shall be two breakaway indicators. One shall be located above the TC indicator with the second indicator located above the RA indicator as shown in figures 2 and 3. Each indicator shall contain four lamps and shall be aviation red in color with a minimum light output in bright areas of 300 foot-lamberts. The indicators shall be illuminated by a +28 VDC signal on connector pin P1-97. See table VII, input 9. Each indicator shall be capable of being dimmed by the pilot's control panel dimmer switch.

3.4.13 Inputs. The inputs specified in 3.4.13.1 through 3.4.13.8.8 shall be supplied to the ADI.

3.4.13.1 Inputs from ballistics computer.

3.4.13.1.1 Flight path and command heading lines.

3.4.13.1.1.1 Path on-off. An on-off dc voltage shall be provided on connector pin P1-38. (See 3.4.20.1 and table VII, input 50.)

3.4.13.1.1.2 Path vanishing point vertical position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-37 which displaces the apparent vanishing point of the path vertically in a stabilized coordinate system. (See 3.4.20.3.1 and table VII, input 49.)

3.4.13.1.1.3 Path vanishing point horizontal position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-36 which displaces the apparent vanishing point of the path horizontally in a roll stabilized coordinate system. (See 3.4.20.3.2 and table VII, input 48.)

3.4.13.1.1.4 Path near-end position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-40 which displaces the path near-end horizontally. See table VII, input 52. Horizontal positioning shall not be given during the attack and navigation modes and the voltage shall remain at the reference zero VDC position. (See 3.4.20.3.3.)

3.4.13.1.1.5 Path near-end change with altitude. An analog dc voltage (zero to +15 V) shall be provided on connector pin P1-41 which spreads the edges of the path near-end in a roll stabilized coordinate system. (See 3.4.20.3.4 and table VII, input 53.)

3.4.13.1.1.6 Command heading lines motion. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-39 which causes each Command Heading Line (CHL) to move horizontally. See table VII, input 51. Plus or minus 15 volts corresponds to maximum velocity of the CHL. (See 3.4.21.2.1.)

3.4.13.1.2 Pull-up marker.

3.4.13.1.2.1 Pull-up marker on-off. An on-off dc voltage shall be provided on connector pin P1-42. (See 3.4.17.5.1 and table VII, input 54.)

3.4.13.1.2.2 Pull-up marker position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-43 which shall be a function of the distance to the pull-up point. (See 3.4.17.5.2 and table VII, input 55.)

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3.4.13.1.2.3 Pull-up path radius of curvature. An analog dc voltage (zero to +15 V) shall be provided on connector pin P1-44 which serves to vary the apparent curvature of the pull-up path with respect to "g" level. (See 3.4.17.5.3 and table VII, input 56.)

3.4.13.1.3 Release marker.

3.4.13.1.3.1 Release marker on-off. An on-off dc voltage shall be provided on connector pin P1-45. (See 3.4.17.4.1 and table VII, input 57.)

3.4.13.1.3.2 Release marker position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-46 to vertically position the release marker in both straight and curved path attack situations. (See 3.4.17.4.2 and table VII, input 58.)

3.4.13.1.3.3 Release marker size. An analog dc voltage (zero to +15 V) shall be provided on connector pin P1-47 which shall control the size of the release marker so that it will appear in proper perspective. (See 3.4.17.4.3 and table VII, input 59.)

3.4.13.1.4 Impact point.

3.4.13.1.4.1 Impact point horizontal position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-54 which shall be proportional to the horizontal displacement in a roll stabilized coordinate system. (See 3.4.17.2.1 and table VII, input 65.)

3.4.13.1.4.2 Impact point vertical position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-53 which shall be proportional to the vertical displacement in a roll stabilized coordinate system. (See 3.4.17.2.2 and table VII, input 64.)

3.4.13.1.4.3 Stall signal. A +28 VDC signal shall be provided on connector pin P1-63 when the aircraft approaches its stall angle of attack. (See 3.4.17.2.3 and table VII, input 69.)

3.4.13.1.5 Weapon symbol.

3.4.13.1.5.1 Weapon symbol on-off. An on-off dc voltage shall be provided on connector pin P1-60. See 3.4.17.6.1 and table VII, input 66. Circuits shall be incorporated so that a second source of the weapon symbol on signal can override the off signal. (See 3.4.13.2.1.)

3.4.13.1.5.2 Weapon symbol horizontal position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-61 which is proportional to the azimuth angle to the target plus the weapon position error signal in azimuth. (See 3.4.17.6.2 and table VII, input 67.)

3.4.13.1.5.3 Weapon symbol vertical position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-62 which is proportional to the elevation angle to the target plus the weapon position error signals in elevation. (See 3.4.17.6.3 and table VII, input 68.)

3.4.13.1.6 Target symbol.

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3.4.13.1.6.1 Target symbol on-off. An on-off dc voltage shall be provided on connector pin P1-50. (See 3.4.17.3.1 and table VII, input 60.)

3.4.13.1.6.2 Target symbol horizontal position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-52 which shall move the target symbol horizontally as a function of the horizontal displacement of the target, in a roll stabilized coordinate system. (See 3.4.17.3.2 and table VII, input 63.)

3.4.13.1.6.3 Target symbol vertical position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-51 which shall move the target symbol vertically as a function of the vertical displacement of the target, in a roll stabilized coordinate system. (See 3.4.17.3.3 and table VII, input 62.)

3.4.13.1.7 Computer reliability signal. A +28 VDC signal shall be provided to indicate improper operation of the computer or when the computer is in test or off. (See 3.4.7 and table VII, input 47.)

3.4.13.1.8 Command over-the-shoulder pull-up signal. A +28 VDC signal shall be provided on connector pin P1-67 which shall indicate that the pilot is being commanded to follow an over-the-shoulder maneuver. This signal shall be applied to the roll commutator switch. (See 3.4.5.1 and table VII, input 73.)

3.4.13.1.9 Vertical speed indicator position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P2-21 and P2-19 which shall move the Vertical Speed Indicator (VSI) marker vertically as a function of the vertical velocity of the aircraft. (See 3.4.28.4 and table VII, inputs 85 and 86.)

3.4.13.2 Inputs from Bullpup Automatic Guidance System (BAGS).

3.4.13.2.1 BAGS weapon symbol on-off. An on-off dc voltage shall be provided on connector pin P1-29. The on signal (+28 VDC) shall result in the display of the weapon symbol and shall be capable of overriding the off signal from the computer. (See 3.4.13.1.5.1). This signal shall also switch the weapon symbol deflection circuits to accept BAGS deflection signals instead of ballistics computer deflection signals. (See table VII, input 14.)

3.4.13.2.2 BAGS weapons symbol horizontal position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-74. (See table VII, input 11.) The horizontal positioning voltages in BULLPUP mode shall have the same dynamic range as specified in 3.4.17.6.2.

3.4.13.2.3 BAGS weapon symbol vertical position. An analog dc voltage (-15 V to +15 V) shall be provided on connector pin P1-75. (See table VII, input 12.) The vertical positioning voltages in BULLPUP mode shall have the same dynamic range as specified in 3.4.17.6.3.

3.4.13.3 Inputs from INS.

3.4.13.3.1 Platform reliability signal. A +28 VDC signal shall be provided on connector pin P2-50 to indicate loss of attitude input information. (See 3.4.6 and table VII, input 43.)

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3.4.13.3.2 Roll angle. The stator of a synchro transmitter (115 V 400 Hz RS911-6A Kearfott or equivalent) located in the INS shall supply on connector pins P2-24, P2-25, P2-26 roll information to the stator of the synchro transformer (11.8 V 400 Hz R1001-1A Kearfott or equivalent) located in the ADI. (See table VII, inputs 37, 38, and 39.)

3.4.13.3.3 Elevation angle. Elevation angle information from a potentiometer (700-1203 Spectrol or equivalent) located in the the INS shall be provided to the ADI on connector pins P2-4 and P2-5. (See 3.4.14.2.1 and table VII, input 34, and input 35.)

3.4.13.3.4 Azimuth angle. Azimuth angle information from a resolver (3R982-002-0 Kearfott or equivalent) located in the INS shall be provided to the ADI on connector pins P2-35 and P2-37. (See 3.4.14.2.2 and table VII, inputs 40 and 41.)

3.4.13.3.5 INS fail safe. Two +28 VDC signals shall be provided on connector pins P2-51 and P2-56. Loss of either signal due to failure of the +28 VDC (filtered) or +28 VDC (unfiltered) signal shall result in CRT blanking. (See table VII, inputs 44, 45, and 46.)

3.4.13.4 Inputs from pilot's control panel.

3.4.13.4.1 System mode of operation. The aircraft shall provide the AC and DC voltages necessary to activate the ADI in the following modes of operation:

a. Switch 1: Off
Standby
Contact analog
Terrain clearance

b. Switch 2: Test

3.4.13.4.2 Standby mode. During standby mode of operation, rated voltage shall be applied to the ADI but not necessarily to any components.

3.4.13.4.3 Contact analog mode. During CA mode of operation, the information shall be capable of being as specified in table I.

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TABLE I. Symbology/mode characteristics.

GROUP 1 VIDEO	GROUP 2 VIDEO
Impact point Target symbol Weapon symbol Pull-up marker Release marker Path border and centerline Pitch lines (+90°, +60°, +30°, -30°, -40°) Roll pointer Command heading lines True/magnetic heading scale Angle of attack reference and error symbol Radar altitude thermometer and numerics Radar altitude/VSI scale ALS/ILS steering VSI marker ALS or ILS alphanumerics	Horizon line Ground texture Pitch trim reference cursors Pitch lines (+10°, +20°, -10°, -20°) Flight path Sky plane Ground plane

3.4.13.4.3.1 Pitch trim cursor input. When input 33(b) is present, input 33(a) shall displace the pitch trim cursor only. When input 33(b) is absent, input 33(a) shall displace the pitch trim cursor and the off-set impact point marker. (See table VII.)

3.4.13.4.4 Terrain clearance mode. During TC mode of operation, group 1 video information of 3.4.13.4.3 plus the offset impact point marker shall be capable of being displayed. If input 33(b) is present, the pitch trim cursor shall also be capable of being displayed.

3.4.13.4.5 Test mode. Test position, switch 2, which is not interlocked with the four-position switch 1 (see 3.4.13.4.1), shall provide the dc voltages on connector pins D1-18 and D1-19 necessary to display the test pattern shown in figure 4. (See table VII, inputs 25 and 26.)

3.4.13.4.6 Dimming capability. The aircraft dimmer switch shall have the capability of dimming the computer reliability indicator (see 3.4.7), the search radar terrain clearance reliability indicator (see 3.4.8), the altimeter reliability indicator (see 3.4.9), the RNG/LASER indicator (see 3.4.10), the attack indicator (see 3.4.11) and the two breakaway indicators (see 3.4.12). The inertial subsystem reliability indicator (see 3.4.6) shall not have the capability of being dimmed.

3.4.13.5 Input from flight mode control panel.

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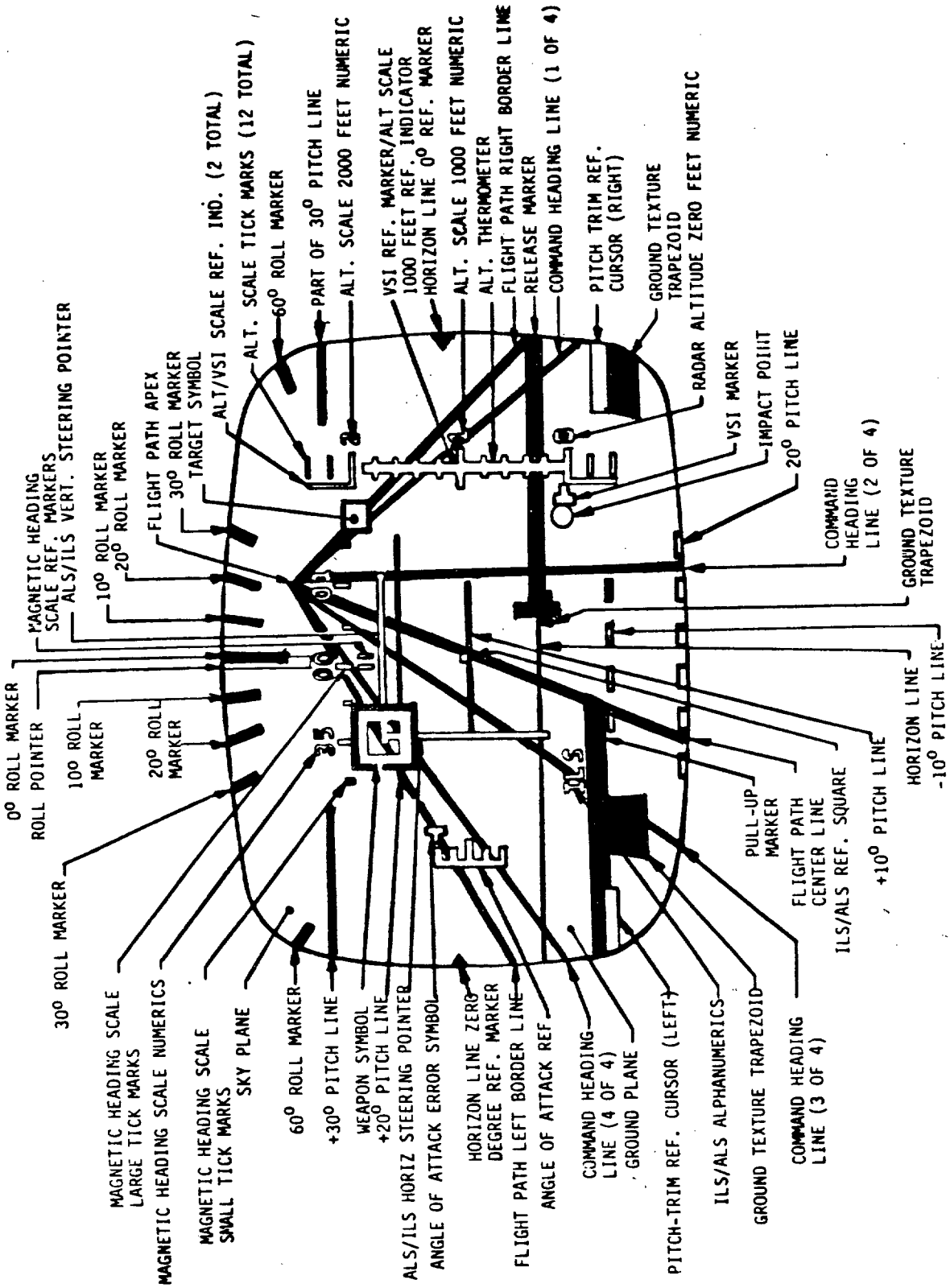


FIGURE 4. Test Mode.

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3.4.13.5.1 Take-off and landing modes. When the switch, identified in 3.4.13.4.1, is in the TC or CA position and the airplane flight mode switch on the flight mode control panel is in the take-off or landing mode, zero VDC shall be provided on connector pin P1-35, to effect the presence of the flight path centerline and the removal of the command heading lines. Conversely, when not in take-off or landing mode, 28 VDC shall be provided to effect the presence of the command heading lines and the removal of the flight path centerline. (See table VII, input 2.)

3.4.13.6 Inputs from RDC.

3.4.13.6.1 Video. Video shall be provided on connector and P2-48 from the RDC by means of a 93 ohm line. The signal shall have a maximum amplitude of 1.6 V_{pp}, black negative. (See table VII, input 4.)

3.4.13.6.2 Altimeter reliability signal. A +28 VDC signal shall illuminate the altimeter reliability indicator to indicate loss of useable altitude information to the RDC. (See 3.4.9.) This signal, when present, shall also blank the radar altitude thermometer and numerics. (See 3.4.13.8.4.1.)

3.4.13.6.3 Search radar reliability signal. A +28 VDC signal shall illuminate the search radar terrain clearance reliability indicator to indicate loss of useable search radar information to the RDC. (See 3.4.8.)

3.4.13.7 Inputs from TRAM.

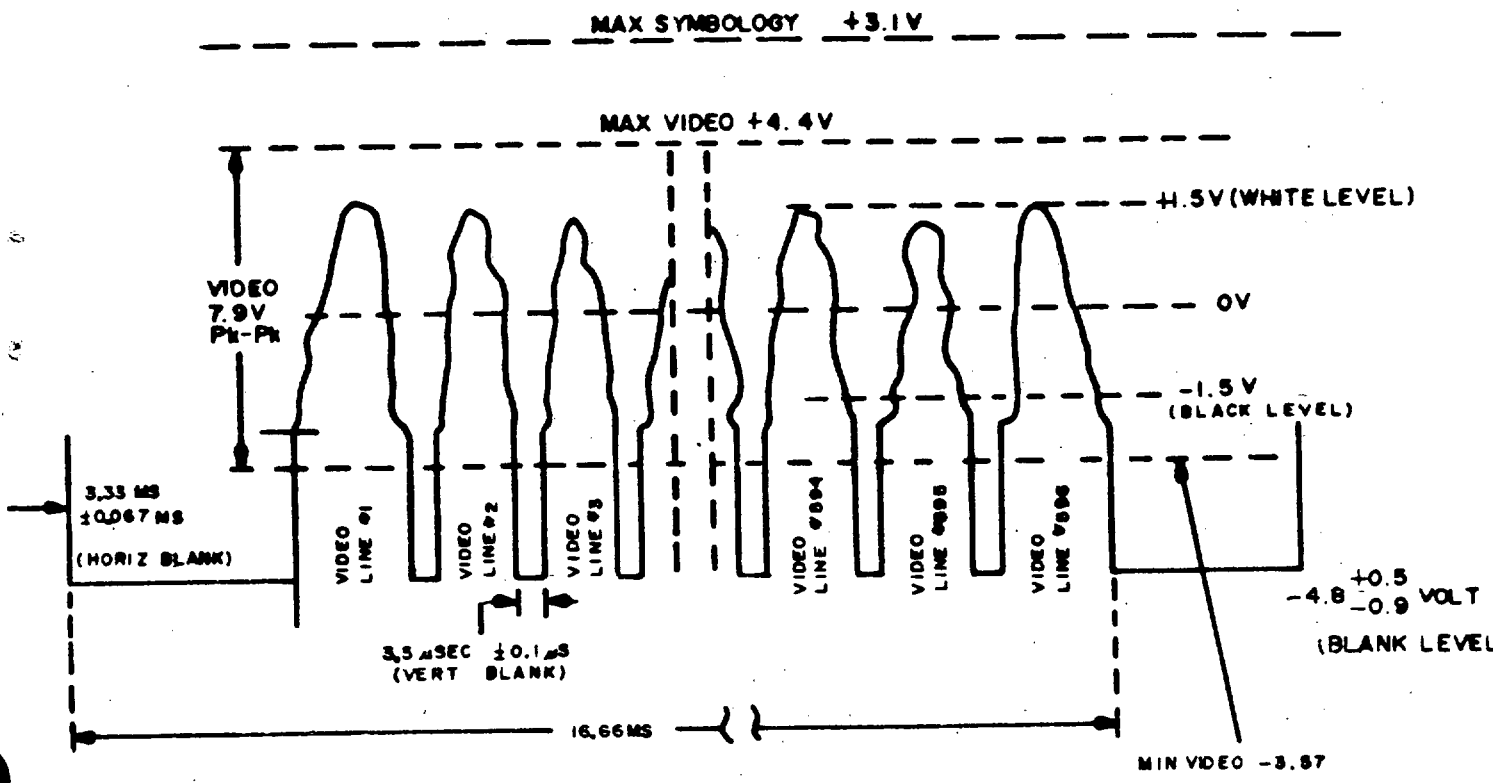
3.4.13.7.1 FLIR inputs. The Detecting and Ranging Set (DRS), AN/AAS-33A, shall provide FLIR video and the associated timing signals to the ADI. Characteristics of these signals are shown in figures 5 and 6.

3.4.13.7.1.1 FLIR video. FLIR video is provided on connector pin P2-13. Characteristics of the FLIR video signal are described below and in figure 5. (See table VII, input 107.)

a. Video content between +4.4 V and -3.5 V (+ is white, - is black).
b. Blanking level is -4.8 V, +0.5 -0.9.
c. Vertical retrace blanking between video lines is 3.5 usec ±0.1 usec.
d. Horizontal retrace blanking is 3.333 msec ±0.067 msec.
e. 596 video lines between horizontal retrace blanking.
f. Input impedance is 75 ohms ± 5 percent (differential amplifier input).

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(A) FLIR horizontal rate



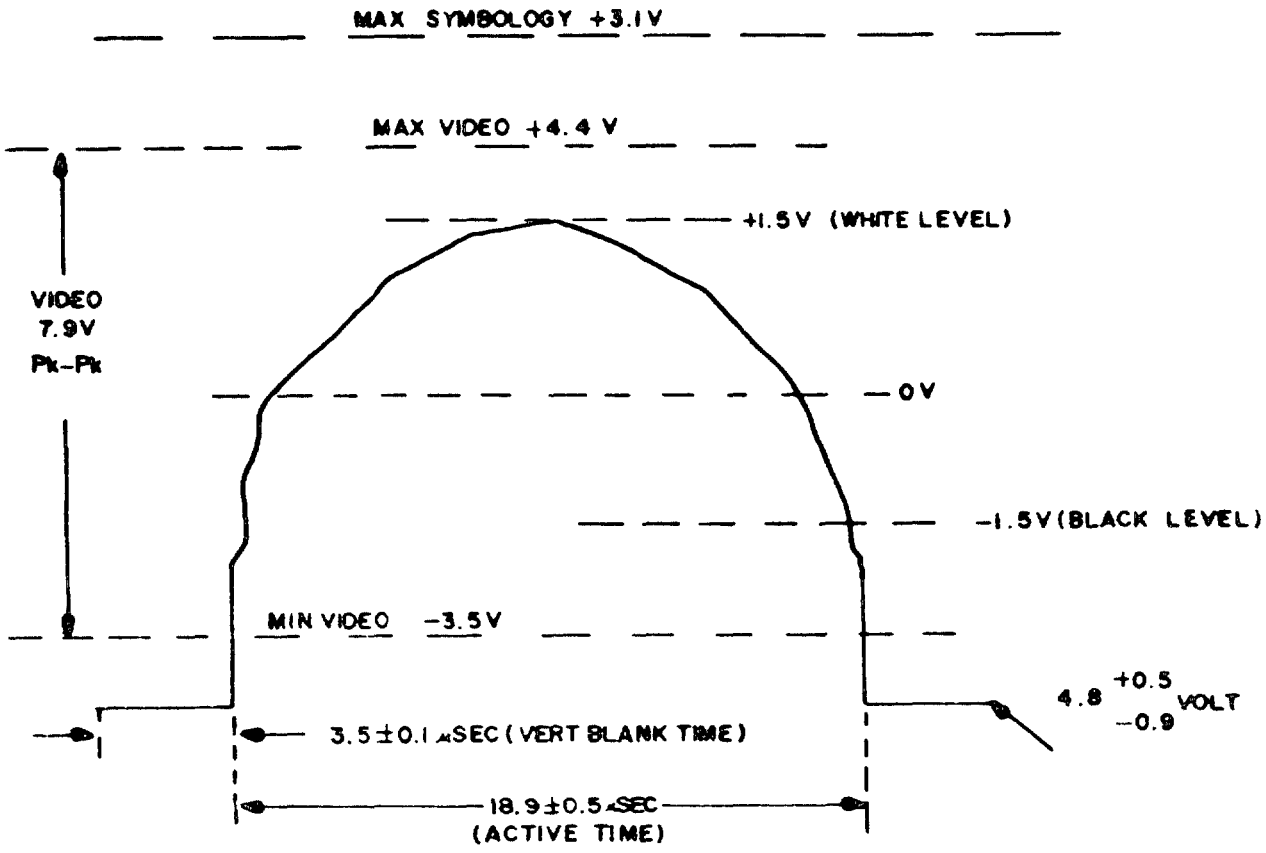
NOTES:

1. Horizontal blanking pulse rise and fall time to be $\leq$ 50 msec.
2. Video bandwidth - 6.0 MHz.
3. Video amplitude can be varied from 0 to 7.9 V_{pp} with the FLIR gain control. Nominal level is 0.5 to 1.0 V_{pp} about 0V.
4. Symbology output rides on top of white video level (variable 0 to 3.1 V).

FIGURE 5. FLIR Video.

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(B) FLIR vertical rate

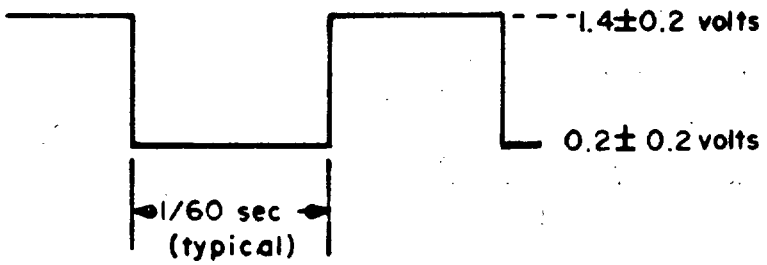


NOTES:
1. Vertical blanking pulse rise and fall time to be $\leq$ 50 nsec.
2. Video bandwidth - 6.0 MHz.
3. Video amplitude can be varied from 0 to 7.9 V_{pp} with the FLIR gain control. Nominal level is 0.5 to 1.0 Vpp about 0 V.
4. Symbology output rides on top of white video level (variable 0 to 3.1 V).

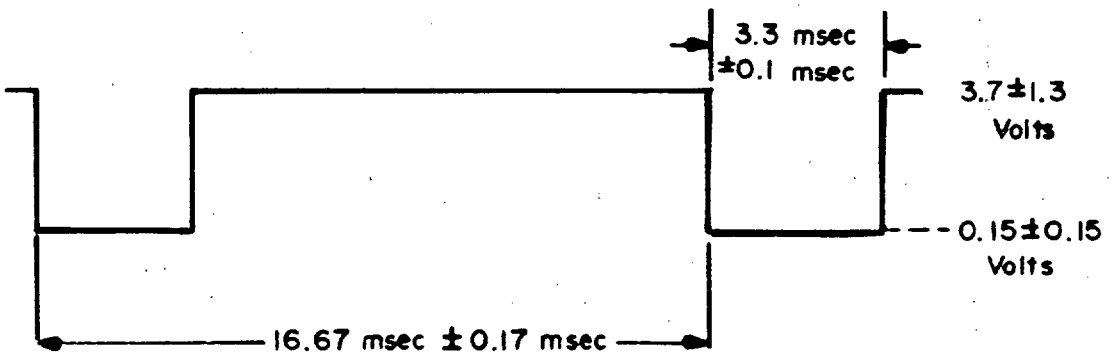
FIGURE 5. FLIR video. - Continued

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(A) FLIR interlace



(B) Horizontal sync



(C) Vertical sync

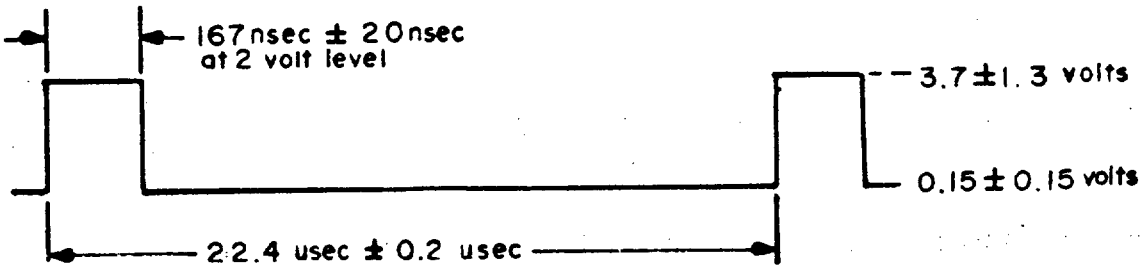


FIGURE 6. FLIR signals.

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3.4.13.7.1.2 FLIR interlace. FLIR interlace is provided on connector pins P2-31 and P2-32. This signal biases the raster vertically on the display. Characteristics of this signal are listed below and are shown in figure 6(A). (See table VII, inputs 108 and 109.)

a. Square wave steps, from 0 to 1.4 V.
b. Maximum rise and fall time is 100 usec.
c. Vertical sweep is from top to bottom.
d. A positive input voltage biases the display raster down.
e. 1.4 V causes displacement equal to 1/224 times the vertical raster size.
f. Input impedance is 75 ohms ± 5 percent (differential amplifier input).

3.4.13.7.1.3 Horizontal sync. Horizontal sync is provided on connector pin P2-49. Characteristics of the horizontal sync signal are described below and in figure 6(B). (See table VII, input 94.)

a. Maximum rise and fall times are 0.1 usec.
b. Leading edge of positive going interval shall occur at the start of horizontal scan (end of blank).
c. Horizontal scan is from left to right.
d. Input impedance is 130 ± 10 ohms (differential amplifier input).

3.4.13.7.1.4 Vertical sync. Vertical sync is provided on connector pin P2-45. Characteristics of the vertical sync signal are described below and in figure 6(C). (See table VII, input 93.)

a. Maximum rise and fall times are 40 nanoseconds.
b. Vertical retrace blanking starts 167 nanoseconds after leading edge of vertical sync pulse.
c. Vertical sweep is from top to bottom.
d. Input impedance is 130 ± 10 ohms (differential amplifier input).

3.4.13.7.2 External video. External video shall be provided to the ADI on connector pin P2-41. Characteristics of this video are shown in figure 1 of MIL-T-81624. (See table VII, input 91.)

3.4.13.7.3 TRAM modes.

3.4.13.7.3.1 FLIR mode on-off. Activation of the FLIR mode is made remotely, at the pilot's discretion. A +28 VDC discrete (TRAM on-off) shall be provided to the ADI on connector pin P2-38 to initiate FLIR mode. The external mode discrete shall be off. (See 3.4.13.7.3.2 and table VII, inputs 95 and 96.) Upon receipt of this signal, the following shall occur:

a. CA or TC modes as described (See 3.4.13.4.3 and 3.4.13.4.4) shall be interrupted within the ADI.
b. The yoke shall be rotated clockwise to the 90 degree position and locked in this position to prevent double roll of the FLIR presentation. The rolling of the yoke shall enable the high speed line scan circuitry of the ADI to accommodate the high speed vertical sweep requirement of the FLIR raster. The ADI lower speed field sweep circuitry shall accomodate the low frequency horizontal sweep requirement of the FLIR raster.

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c. FLIR video shall be displayed with an aspect ratio of four units horizontal by three units vertical.
d. Symbology generated within the ADI shall not be displayed.

3.4.13.7.3.2 External mode on-off. External mode on-off shall be provided to the ADI on connector pin P2-39. (See table VII, input 96.) External mode shall be activated when both the TRAM mode on-off discrete and the external mode +28 VDC on-off discrete are present. (See 3.4.13.7.3.1.) The presence of the external mode discrete alone shall have no effect on CA and TC presentations.

3.4.13.7.3.2.1 External mode. Upon activation of the external mode, the following shall occur:

a. CA and TC modes shall be interrupted within the ADI. (See 3.4.13.4.3 and 3.4.13.4.4.)
b. Raster shall be locked to zero degrees, to prevent double roll of the external video presentation. The currents producing the low frequency sweep shall be reversed through the yoke to provide the proper scan direction.
c. External video shall be displayed with an aspect ratio of four units horizontal by three units vertical.
d. Symbology generated within the ADI shall not be displayed.

3.4.13.8 Additional symbol inputs.

3.4.13.8.1 True/magnetic heading scale. True or magnetic heading data shall be provided to the ADI on connector pins P2-27, P2-28, and P2-29 in the form of a 3 wire synchro transmitter output. See table VII, inputs 97, 98, and 99. The synchro transmitter shall be a CM41011020 or HGC8LN06L600. The synchro transmitter shall be excited by an external 2 wire 26 VAC single phase source.

a.	Heading range	0 to 360°
b.	Voltage range	±11.8 VAC, 400 Hz, single phase
c.	Zero degree reference	Zero ARINC high null
d.	Load	Synchro receiver
e.	Reference voltage	Primary 115 VAC, 400 Hz, phase 1.

3.4.13.8.2 Angle of attack. The Angle Of Attack (AOA) input signal shall be provided from an AOA probe on connector pins P2-1 and P2-30 with the characteristics listed below. (See table VII, inputs 101 and 100.)

a.	Output voltage range	0 to +10.0 VDC
b.	Scale factor	10.0 VDC is 30 units 0.0 VDC is 0 units
c.	Maximum rate of change	50°/sec
d.	Minimum ADI input Z	100 K ohms
e.	Probe output circuit	2700 ohm potentiometer (±5 percent).

3.4.13.8.3 Landing mode signal. A landing mode signal shall be provided on connector pin P2-18 and shall be used to enable the AOA symbol. The symbol shall be displayed when +28 VDC is present. The symbol shall not be displayed when zero VDC (open circuit) is present. (See table VII, input 112.)

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3.4.13.8.4 Radar altitude. Radar altitude information shall be provided directly from the APN-141/194 system on connector pins P2-2 and P2-33. (See table VII, inputs 102 and 111.) The signal characteristics are shown below:

a.	Voltage range	+1.0 to +25.0 VDC
b.	Altitude range	0 to 5000 feet
c.	Maximum rate of change	3000 ft/sec
d.	Scale factor, 0-400 ft.	20 mV/ft
e.	Voltage level, 0 ft.	+1.0 VDC
f.	Voltage level, 400 ft.	+9.0 VDC
g.	Voltage level, 5000 ft.	+21.0 VDC
h.	Maximum voltage	+25.0 VDC
i.	Minimum ADI input Z	100 K ohms

3.4.13.8.4.1 Altitude reliability signal. The altitude reliability signal shall be used as a validity signal. Zero VDC means the signal is valid; +28 VDC means the signal is invalid. (See 3.4.9, 3.4.13.6.2 and table VII, input 5.)

3.4.13.8.5 ACLS/ILS steering. Steering information shall be provided by either automatic carrier landing system (ACLS) or instrument landing system (ILS) on connector and pins P2-8 and P2-14. (See table VII, inputs 103 and 104.) (See table II.)

TABLE II. ACLS/ILS signal characteristics.

Limits	Horizontal input	Vertical input
Voltage range	±20 VDC	±20 VDC
Full scale voltage	±4.4 VDC	±4.4 VDC
Minimum ADI input Z	100 K ohms	100 K ohms
Polarity (+VDC)	Right	Up

3.4.13.8.6 Validity signal. A validity signal shall be provided on connector pin P2-15. A +28 VDC signal means the steering information is valid and the ACLS/ILS steering symbol should be displayed. When the validity signal indicates valid, the target symbol, weapon symbol, command heading lines, path, path border, and centerline shall not be displayed. (See table VII, input 105.)

3.4.13.8.7 Mode signal. In addition to the ACLS/ILS validity signal, a mode signal shall be provided on connector pin P2-20 to indicate that the source of steering information is either ACLS or ILS. Zero VDC shall cause the letters "ALS" to appear on the display if the validity signal is also present. A +28 VDC signal shall cause the letters "ILS" to appear on the display if the validity signal is present. If no validity signal is present neither set of letters shall appear on the display. (See table VII, input 113.)

3.4.13.8.8 Added symbology capability. Inputs 114 and 115, represented by connector pins P2-43 and P2-44, may be connected together in the aircraft and shall provide a maximum impedance of 10 ohms to indicate that the AOA, ACLS/ILS steering, radar altitude, VSI, and true/ magnetic heading symbols may be displayed. An open circuit between pins P2-43 and P2-44 shall indicate that the above symbols shall not to be displayed. (See table VII, inputs 114 and 115.)

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3.4.14 Outputs. The following outputs shall be provided by the ADI.

3.4.14.1 Sync outputs. Vertical and horizontal sync pulses shall be provided to the ADI on connector pins P2-46 and P2-47. (See table VIII, outputs 4 and 5.) The sync characteristics are shown in table III.

TABLE III. Output Sync.

Characteristic	Horizontal sync pulse	Vertical sync pulse
Repetition rate	15,750 Hz	60 Hz
Amplitude	5 V ± 10 percent	5 V ± 10 percent
Pulse width	12 ± 1.0 usec (measured at 50 percent maximum amplitude points)	500 ± 50 usec (measured at 50 percent maximum amplitude points)
Rise time	1.0 usec maximum (between 10 percent and 90 percent points)	3 usec maximum (between 10 percent and 90 percent points)
Fall time	1.0 usec maximum (between 10 percent and 90 percent points)	12 usec maximum (between 10 percent and 90 percent points)
Termination	90 to 220 ohms (direct couple to ground)	90 to 330 ohms (direct couple to ground)

3.4.14.2 Outputs to INS.

3.4.14.2.1 Elevation angle. The ADI shall provide on connector pins P2-6 and P2-7, the necessary excitation for the elevation angle potentiometer located in the INS adapter unit. (See 3.4.13.3.3 and table VIII, outputs 6 and 7.)

3.4.14.2.2 Azimuth angle. The ADI shall provide on connector pin P2-17 the necessary 31,500 Hz excitation voltage for the azimuth resolver located in the INS adapter unit. (See 3.4.13.3.4 and table VII, output 8.)

3.4.14.2.3 Synchro transmitter excitation. The ADI shall provide on connector pin P2-22 a 115 V, 400 Hz phase 1 excitation voltage for the synchro transmitter located in the INS. (See 3.4.13.3.2 and table VIII, output 9.)

3.4.15 General display requirements. The ADI shall provide a useable display area of not less than 5.2 inches in the vertical direction by a minimum of 7.0 inches in the horizontal direction. These dimensions shall result in the display of real world angular dimension of approximately 60° in the vertical by approximately 50° in the horizontal. The total raster (including overscan) shall be a square. The edges of the raster shall not become visible at any roll angle.

3.4.15.1 Display center in CA and TC modes.

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3.4.15.1.1 Raster vertical position. By rotating the raster through 360°, the raster vertical center, defined as a virtual point delayed 8.73 msec from the leading edge of the vertical sync signal, shall pivot on one point on the CRT face. This point shall be the display center.

3.4.15.1.2 Raster horizontal position. By rotating the raster through 360°, the raster horizontal center, defined as a virtual point delayed 39.25 usec from the leading edge of the horizontal sync signal, shall pivot on one point on the CRT face. This point shall be the display center.

3.4.15.2 Display brightness.

3.4.15.2.1 Brightness control. Display brightness shall be continuously variable by use of a manual control located on the pilot's control panel. The manual control shall vary the display brightness continuously from maximum useable brightness to fully blanked. The ADI shall produce a display of such brightness that it shall be capable of use in the cockpit without the aid of a hood or shade under a daylight 10,000 foot-lamberts solar ambient condition.

3.4.15.3 Display resolution and contrast.

3.4.15.3.1 Resolution. The ADI shall exhibit as a minimum the horizontal and vertical display resolution shown in table IV. The required resolution values are expressed in active written lines per inch. The resolution values apply over the central 80 percent of the display area, and the resolution outside this area may be up to 25 percent less than specified herein.

TABLE IV. ADI resolution.

Operation mode	Resolution in lines per inch	
	Vertical	Horiztonal
CA and TC	60	61
Tram-External	60	79
Tram-FLIR	60	78

3.4.15.3.2 Contrast range of the display. A linear relationship shall exist between voltage excursions of input signals and the corresponding luminance on the face of the CRT. This linear relationship shall be maintained throughout the range of signal input and shall provide at least six distinct steps of a gray scale between the extreme limits of luminance on the CRT. Circuitry shall be included, if required, to maintain this linear relationship. If corrective circuitry is used, extreme operating conditions should be taken into account to allow for day and night operation.

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3.4.15.3.3.1 Sky plane. The ADI shall present a sky plane which has a contrast which is readily discernable from the horizon and ground plane.

3.4.16 Ground texture generator. The ground texture generator shall present a quasi-random texture of trapezoids. (See figure 7.)



FIGURE 7. Ground texture.

3.4.16.1 Aspect ratio. The size of the trapezoids shall vary with their apparent depression angle below the horizon according to the approximate equations of perspective. The size of the ground texture elements shall be as specified in 3.4.16.5.

3.4.16.2 Linear velocity. The trapezoids shall move at a linear velocity across the CRT face.

3.4.16.3 Packing density. There shall always be at least three ground texture elements visible at zero degree pitch. There shall always be at least one element visible at -90° pitch.

3.4.16.4 Continuous movement. The movement of the ground texture elements shall be smooth and at a rate of approximately 0.25 inches per second with no apparent accelerations or decelerations. All elements shall appear to emerge from the horizon.

3.4.16.5 Change of element size. The solid angle formed by imaginarily extending the sides of a ground texture element shall continuously, and with no apparent abrupt changes, increase in size as it moves away from the horizon.

3.4.16.6 Lateral movement. The ground texture elements shall move laterally in response to input 40, or connector pin P2-35.

3.4.17 Symbol generators. The ADI shall contain symbol generators.

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3.4.17.1 Zenith marker generator. The zenith marker generator shall provide a +90° pitch reference. The marker shall be a black dashed line consisting of segments 1.0 ± 0.3 inches long. Each segment shall be rectangular and have a vertical dimension of 0.13 ± 0.06 inches as measured with the line located at the vertical center of the display (+90° pitch indication). The separation between segments shall be 1.0 ± 0.3 inches. (See figure 8.) The zenith marker shall not overlay any part of the altitude/VSI scale, heading scale, AOA or ACLS/ILS steering symbols.

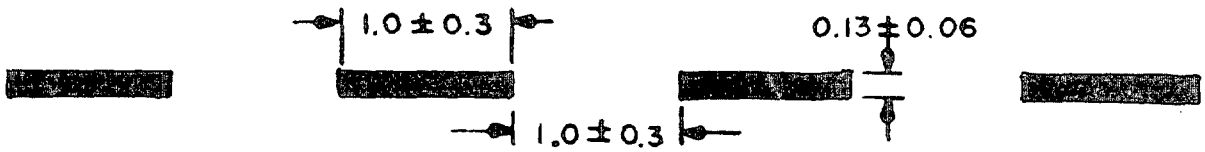


FIGURE 8. +90° zenith marker.

3.4.17.1.1 Zenith marker vertical motion. The zenith marker shall be parallel to the horizontal raster lines for all conditions. All movements of the marker (with respect to the raster) shall be normal to the raster lines and shall indicate real world pitch within ± 0.2 inches ± 0.08 D, where D is the vertical distance in inches from display center to the correct position of the zenith marker center.

3.4.17.2 Impact point generator. The impact point generator shall provide a bright approximately circular marker 0.20 ± 0.06 inches in diameter as measured at the center of the display area. The circle shall be outlined in black to a diameter of 0.27 ± 0.06 inches. The impact point shall indicate the direction of the airplane velocity vector. (See figure 9.)

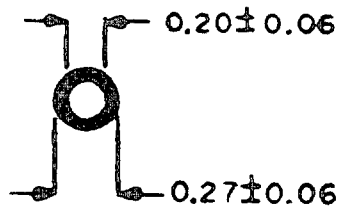


FIGURE 9. Impact point.

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3.4.17.2.1 Impact point horizontal displacement. The impact point shall be displaceable parallel to the horizon. An input signal of + 15 VDC shall move the impact point to a position such that the symbol center is 3.25 ± 0.35 inches to the left of the display center. An input signal of -15 VDC shall move the symbol 3.25 ± 0.35 inches to the right of display center input signals between +15 VDC and -15 VDC shall position the symbol proportionally across the screen. In addition, with a zero VDC input signal, the center of the impact point shall be within ±0.12 inches of the display center.

3.4.17.2.2 Impact point vertical displacement. The impact point shall be displaceable normal to the horizon. An input signal of +15 VDC shall move the impact point to a position such that the symbol center is 2.35 ± 0.31 inches below the center of the display area. An input signal of -15 VDC shall move the symbol 2.35 ± 0.31 inches above display center. Input signals between +15 VDC and -15 VDC shall position the symbol proportionally. In addition, with a zero VDC input signal, the center of the impact point shall be within ±0.12 inches of the display center.

3.4.17.2.3 Stall signal. When the aircraft approaches the stall angle of attack, a discrete +28 VDC signal shall be sent to the ADI. (See 3.4.13.1.4.3.) When this signal is received, the ADI shall blink the impact point at a rate of 2 ± 1 Hz.

3.4.17.3 Target symbol generator. The target symbol generator shall provide a bright approximately square symbol 0.27 ± .06 inches as measured at the center of the display. The square shall be outlined in black by a square of 0.33 ± 0.06 inches. The target symbol position shall indicate relative target azimuth and elevation information. (See figure 10.)

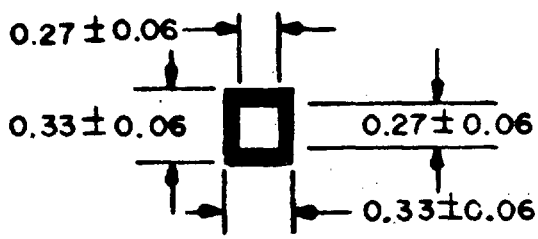


FIGURE 10. Target symbol.

3.4.17.3.1 Target symbol on-off signal. The target symbol shall be biased off when +28 VDC signal is sent to the target symbol generator. When the signal goes to zero VDC (open circuit), the target symbol shall appear on the display. (See 3.4.13.1.6.1.)

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3.4.17.3.2 Target symbol horizontal displacement. The target symbol shall be displaceable parallel to the horizon. An input signal of +15 VDC shall move the target symbol to a position such that the symbol center is 3.25 ± 0.35 inches to the left of the display center. An input of -15 VDC shall move the symbol 3.25 ± 0.35 inches to the right of the display center. Input signals between +15 VDC and -15 VDC shall position the symbol proportionally across the display. In addition, with a zero VDC input signal, the center of the target symbol shall be within ± 0.12 inches of the display center.

3.4.17.3.3 Target symbol vertical displacement. The target symbol shall be displaceable normal to the horizon. An input signal of +15 VDC shall move the target 2.35 ± 0.31 inches below the display center. An input signal of -15 VDC shall move the target 2.35 ± 0.31 inches above the display center. Input signals between +15 VDC and -15 VDC shall position the symbol proportionally across the display. In addition, with a zero VDC input signal, the center of the target symbol shall be within ±0.12 inches of the display center.

3.4.17.4 Release marker generator. The release marker generator shall provide a black release marker which shall be in the form of the letter "T" rotated 90° counterclockwise. The left edge shall be within 0.75 inches maximum to the right of an imaginary vertical center line passing through the display. The horizontal member of the marker shall extend to the right side of the display for all possible vertical locations. Its maximum vertical width shall be 0.25 ± 0.06 inches thick. The maximum dimensions of the vertical member shall be 0.75 ± 0.12 inches high and 0.25 ± 0.06 inches wide. The horizontal member minimum width shall be 0.12 ± 0.06 inches thick. The vertical member minimum dimensions shall be 0.38 ± 0.09 inches high and 0.12 ±0.06 inches wide. These size variations shall be in response to the signal specified in 3.4.13.1.3.3. (See figure 11.)

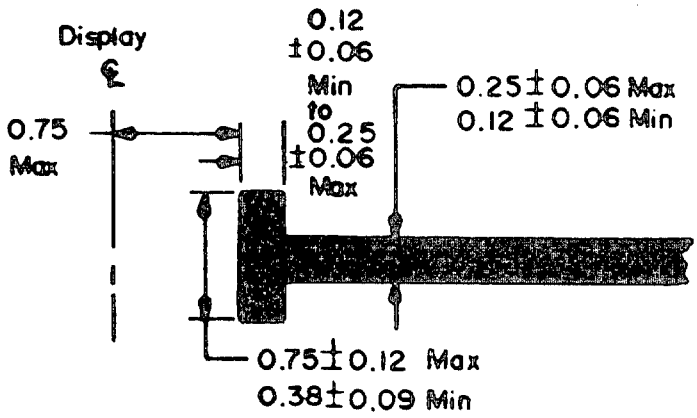


FIGURE 11. Release marker.

3.4.17.4.1 Release marker on-off signal. The release marker shall be biased off when a ±28 VDC signal is sent to the release marker generator. When this signal goes to zero VDC (open circuit), the marker shall appear at a designated position in accordance with 3.4.13.1.3.2 and 3.4.13.1.3.3.

3.4.17.4.2 Release marker symbol position. The horizontal position of the release marker shall be such that it is shown in the right half of the display.

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The vertical position shall be in proportion to the release marker signal specified in 3.4.13.1.3.2. An input signal of -15 VDC shall move the release marker up and under the display mask, while a zero VDC signal shall position the center of the horizontal member within 0.12 inches of the display center. An input signal of +11.8 VDC shall move the release marker to a position such that the center of the horizontal member is 2.48 ± 0.12 inches below the display center and a +15 VDC signal shall move the release marker below the display mask. Input signals between zero and +15 VDC shall position the release marker proportionately on the screen.

3.4.17.4.3 Release marker variable size. The size of the release marker shall be proportional to slant or straight range to the release point. Zero VDC shall correspond to minimum size; a +15 VDC signal shall correspond to maximum size. (See 3.4.13.1.3.3.)

3.4.17.5 Pull-up marker generator. The pull-up marker generator shall provide a black bar which extends horizontally from the left edge of the flight path border to the left edge of the flight path centerline and which grows proportionally in width from 0.12 ± 0.06 inches to 0.25 ± 0.06 inches as it moves down. (See figure 12.)

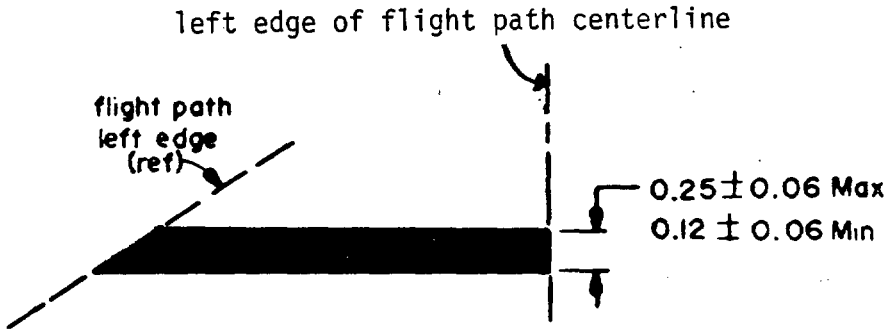


FIGURE 12. Pull-up marker.

3.4.17.5.1 Pull-up marker on-off signal. The pull-up marker shall be biased off when a +28 VDC signal is sent to the pull-up marker generator. When the signal goes to zero VDC (open circuit) the marker shall appear. (See 3.4.13.1.2.1.)

3.4.17.5.2 Pull-up marker symbol position. The horizontal position of the pull-up marker shall be such that it extends from the left flight path line to within ± 0.125 inches of flight path centerline. The vertical position shall be established by the pull-up marker signal as noted in 3.4.13.1.2.2. An input signal of zero Vdc shall position the pull-up marker such that the center of the marker is within ± 0.12 inches of the display center. A ± 7.5 VDC signal shall position the center of the pull-up marker 1.68 ± 0.26 inches below the display

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center. A +15 VDC signal shall position the pull-up marker below the lower edge of the display mask. Position and size limits are not defined in the negative signal input range. Input signals between zero and +15 VDC shall position the symbol proportionally down the display.

3.4.17.5.3 Pull-up radius of curvature. A zero VDC signal shall correspond to 0 "g"; a +15 VDC signal shall correspond to maximum "g". (See 3.4.13.1.2.3.)

3.4.17.6 Weapon symbol generator. The weapon symbol generator shall provide a bright hollow square symbol whose outside dimensions shall be 0.67 ± 0.06 inches and whose edges shall be 0.13 ± 0.06 inches wide with both dimensions measured at the center of the display. The weapon symbol shall have 0.07 ± 0.03 inches black outline. The dimensions of the weapon symbol and target symbol shall be within the ranges specified for each symbol and, in addition, shall be such as to provide a discernible space between the symbols when positioned on a common center. See figure 13. Outside dimensions of the black border shall be 0.80 ± 0.06 inches.

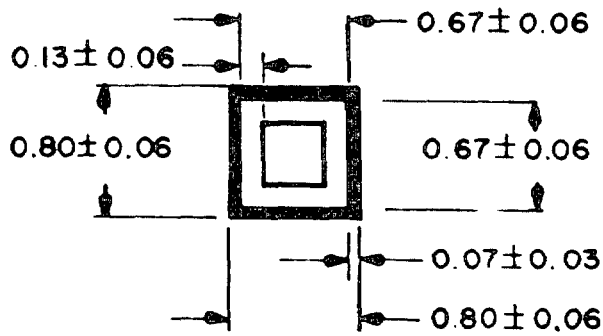


FIGURE 13. Weapon symbol.

3.4.17.6.1 Weapon symbol on-off signal. The weapon symbol shall be biased off when a +28 VDC signal is sent to the weapons symbol generator. When the signal goes to zero VDC (open circuit) the symbol shall appear. (See 3.4.13.1.5.1.)

3.4.17.6.2 Weapon symbol horizontal displacement. In a roll-stabilized coordinate system, an input signal of zero VDC shall position the weapon symbol such that the center of the hollow square is within ± 0.125 inches of the display center. (See 3.4.13.1.5.2.) A +15 VDC input signal shall position the center of the hollow square 3.25 ± 0.35 inches to the left of the display center and a -15 VDC input shall position the center of the weapon symbol 3.25 ± 0.35 inches to the right of the display center. Input signals between +15 VDC and -15 VDC shall position the weapon symbol proportionally across the display.

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3.4.17.6.3 Weapon symbol vertical displacement. In a roll-stabilized coordinate system, an input signal of zero VDC shall position the weapon symbol such that the center of the hollow square is within ±0.125 inches of the display center. (See 3.4.13.1.5.3.) A +15 VDC input signal shall position the center of the hollow square 2.35 ± 0.31 inches below the display center and a -15 VDC input signal shall position the center of the hollow square 2.35 ± 0.31 inches above the display center. Input signals between +15 VDC and -15 VDC shall position the symbol proportionally across the display.

3.4.17.7 Response to a step on-off input. The above symbols shall appear and stabilize at the required position within 250 milliseconds after the ADI receives a step input equal to one half the maximum total excursion of the symbols.

3.4.17.8 Static accuracy. Unless otherwise specified herein, the above symbols shall be positioned as follows where D is the radial distance from the display center.

a. Horizontal positioning shall be ± (0.125 + 0.07 D) inches.
b. Vertical positioning shall be ± (0.125 + 0.08 D) inches.

3.4.18 Pitch line generators. The ADI shall generate +60°, +30°, +20°, +10°, -40°, -30°, -20° and -10° pitch lines. The pitch lines shall not overlay any part of the altitude scale, heading scale, AOA, ACLS/ILS steering or VSI symbols.

3.4.18.1 +60°, +30° and -30° pitch lines. The pitch symbol generators shall provide +60°, +30° and -30° pitch lines. The +60° line shall be 0.13 ± 0.06 inches thick and the +30° and -30° lines shall be 0.07 ± 0.03 inches thick. The pitch lines shall be parallel within ±1° to the horizon with the static accuracy requirements of ± (0.125 ± 0.08D) inches, where D is the pitch line vertical distance in inches from display center as determined by the pitch positioning voltage. The +60°, +30°, and -30° pitch symbols shall be superimposed upon the search radar terrain clearance display. The +30° and +60° pitch lines shall be black and extend completely across the display. The -30° line shall consist of bright segments extending completely across the display. The spaces between segments shall be 0.27 ± 0.06 inches in length. (See figure 14.)

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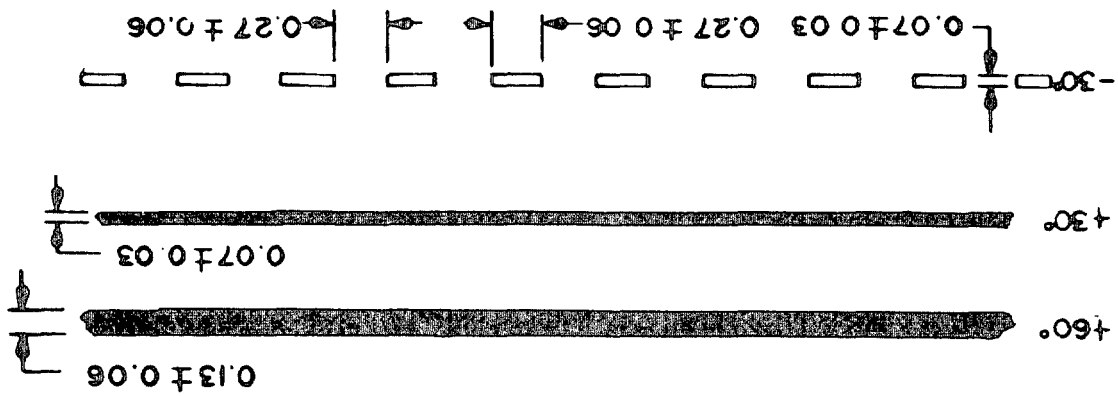


FIGURE 14. +60°, +30° and -30° pitch lines.

3.4.18.2 +20°, +10°, -10° and -20° pitch lines. The pitch symbol generators shall provide +20°, +10°, -10° and -20° pitch lines. The pitch lines shall be 0.07 ± 0.03 inches thick, 1.8 and 2.8 inches in length, respectively and shall be parallel to the horizon. The +10° and +20° pitch lines shall be solid black lines. The -10° and -20° pitch lines shall consist of four and six 0.27 ± 0.06 inches segments, respectively. Spaces shall be equal to segment lengths. The static accuracy of the pitch lines shall be within ± (0.125 ± 0.08D) inches, where D is the vertical distance in inches to the pitch lines from the display center. (See figure 15.)

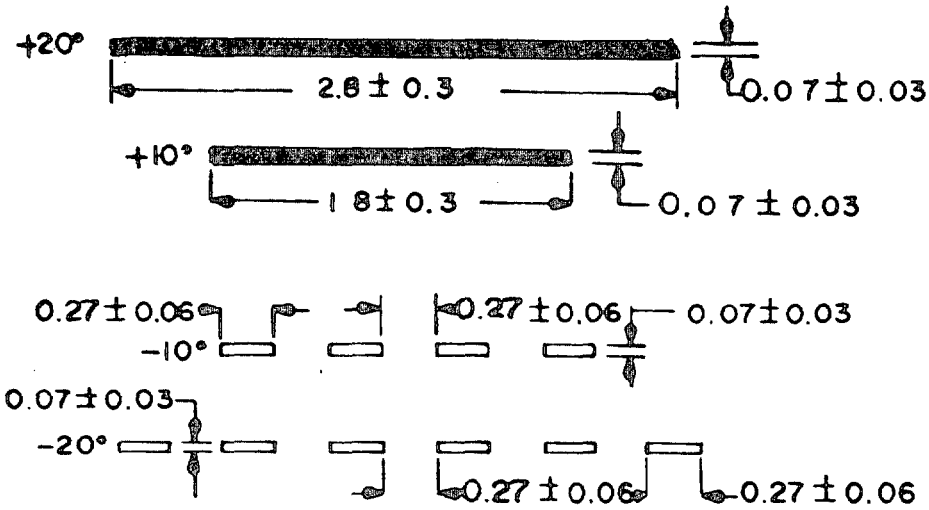


FIGURE 15. +20°, +10°, -10°, and -20° pitch lines.

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3.4.18.3 -40° pitch line. The pitch symbol generator shall provide a segmented bright -40° pitch line, 0.13 ± 0.06 inches thick. The bright segments shall be 0.27 ± 0.06 inches in length. The spaces between segments shall be 0.27 ± 0.06 inches. The -40° line shall be superimposed upon the search radar terrain clearance display. (See figure 16.)

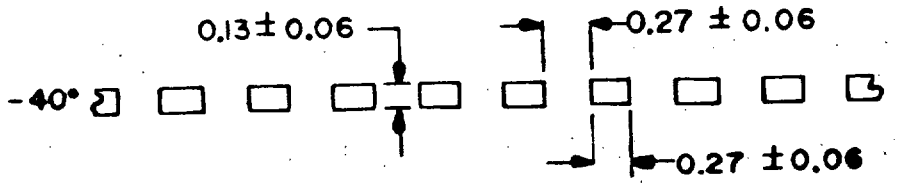


FIGURE 16. -40° pitch line.

3.4.19 Horizon line. The horizon line shall be black. The horizon shall be parallel to the base of the ADI within ±0.5 degree when a zero roll signal is applied to the ADI. The horizon shall correctly follow roll angles to within ±0.5 degree. The horizon line shall be a distance from the display center proportional to the pitch signal in accordance with the static accuracy requirements of ± (0.125 + 0.08D) inches, where D is the vertical distance in inches from display center. The horizon shall stabilize on the correct distance from the center within 250 milliseconds after a 30 degree pitch command is applied to the ADI. The horizon line thickness shall be 0.07 ± 0.03 inches. The horizon line shall not overlay any part of any other symbol. (See figure 17.)

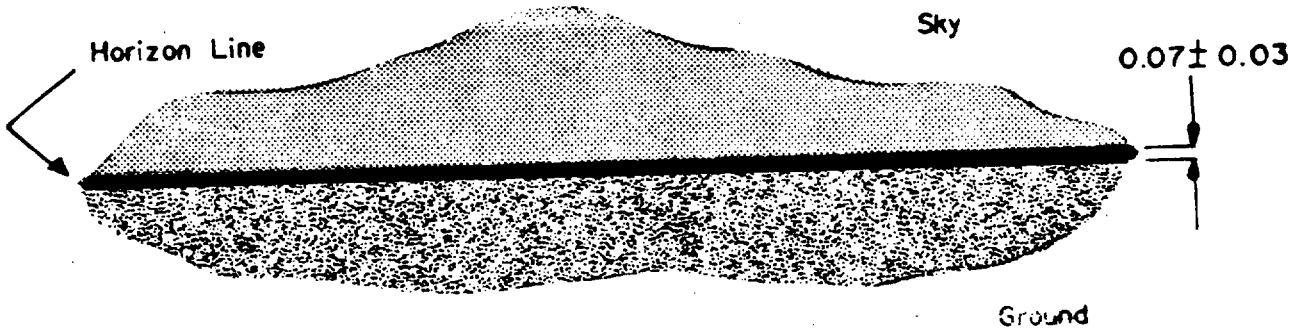


FIGURE 17. Horizon line.

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3.4.20 Flight path generator. The flight path generator shall generate a path which is capable of being superimposed upon the contact analog display. The brightness of the path shall be controllable by means of a path brightness potentiometer (part of the pilot's control panel) which shall vary the path brightness from off (no brightness) to a maximum brightness. The maximum brightness shall be approximately equal to the brightness of the sky plane. (See figure 18.)

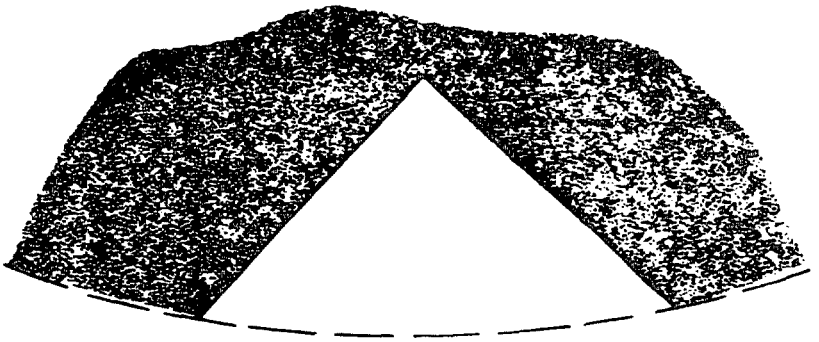


FIGURE 18. Flight path.

3.4.20.1 Path on-off signal. A zero VDC signal shall cause the path to appear on the display; a +28 VDC signal shall cause the path, path border, path centerline, and command heading lines to be blanked. (See 3.4.13.1.1.1.)

3.4.20.2 Path border and centerline. The path shall have a clearly defined black border. A black center line shall be equidistant between the borders, except as specified in 3.4.21. The black borders and centerline shall have a width of 0.12 + 0.06 - 0.08 inches. The path border and centerline shall be capable of being superimposed upon the search radar terrain clearance display. (See figure 19.)

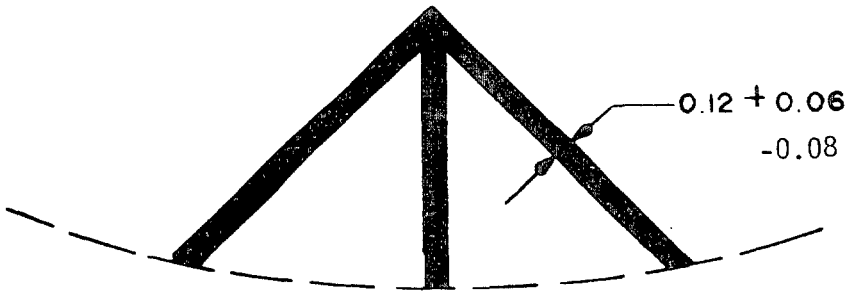


FIGURE 19. Path border and centerline.

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3.4.20.3 Path orientation. The path shall be positioned by the input signal voltages as specified in 3.4.13.1.1.2, 3.4.13.1.1.3, 3.4.13.1.1.4, 3.4.13.1.1.5, 3.4.13.1.2.3, and 3.4.13.1.2.2. Unless otherwise specified herein, the following path positions are specified with a zero degree roll signal and a zero VDC input signal to all path motion inputs other than the one being specified.

3.4.20.3.1 Path apex vertical position. An input signal of +15 VDC shall move the path apex to a point 1.3 ± 0.25 inches below the display center. An input signal of -15 VDC shall move the path apex to a point 1.3 ± 0.25 inches above the display center. An input signal of zero VDC shall move the path apex to the display center ± 0.12 inches.

3.4.21.3.2 Path apex horizontal position. An input signal of +15 VDC shall move the path apex to a point 3.25 ± 0.35 inches to the right of the display center. An input signal of -15 VDC shall move the path apex to a point 3.25 ± 0.35 inches to the left of the display center. An input signal of zero VDC shall move the path apex to the display center ± 0.12 inches.

3.4.20.3.3 Path near-end horizontal position. An input signal of +15 VDC shall position the path near end such that path centerline is 43 ± 4 degrees counterclockwise from a vertical line through the path apex. An input signal of -15 VDC shall position the path near end such that the path centerline is 43 ± 4 degrees clockwise from a vertical line through the path apex. Zero volts DC shall position the path near end such that the path centerline is within ± 3 degrees of a vertical line through the path apex.

3.4.20.3.4 Path near-end size and path symmetry. An input signal of +15 VDC (low altitude) shall position the path edges such that relative to the path centerline the left and right edges are 66 ± 3 degress clockwise and counterclockwise, respectively. An input signal of +7.5 VDC shall position the path edges such that relative to the path centerline, the left and right edges are 52 ± 3 degrees clockwise and counterclockwise, respectively. An input signal of zero volts DC (high altitude) shall position the path edges such that relative to the path centerline the left and right path edges are 31 ± 8 degrees clockwise and counterclockwise, respectively.

3.4.20.3.5 Inverted path. With the inputs necessary to indicate a command over-the-shoulder maneuver, the path shall become inverted and shall have the same size and position (within ± 0.50 inches) as the normal path immediately before inversion. (See 3.4.5.1 and 3.4.13.1.8.)

3.4.20.3.6 Pull-up indication. The path shall appear to rise from the pull-up marker position at a radious of curvature defined by a radius of curvature signal from the ballistic computer. An input signal of zero VDC shall result in no curvature of the path (zero "g"). An input signal of +15 VDC shall result in maximum curvature of the path. If the pull-up marker position input signal is equal to or greater than +9.0 ± 1 VDC, the path near end width control shall be internally switched from the path near end width input signal (command altitude) to a fixed voltage which simulates a path near end width signal of +7.5 VDC. the path apex vertical position input signal shall be internally disconnected and the control of the path apex vertical position shall be switched to the path near end width input signal (command altitude). An input signal (command altitude) of +7.5 VDC and a radius of curvature input signal of +15 vdc shall then cause the path edges to intersect the top of the display mask 1.87 ± 0.25 inches from the center of the path. (See 3.4.13.1.1.8 and 3.4.17.5.2.)

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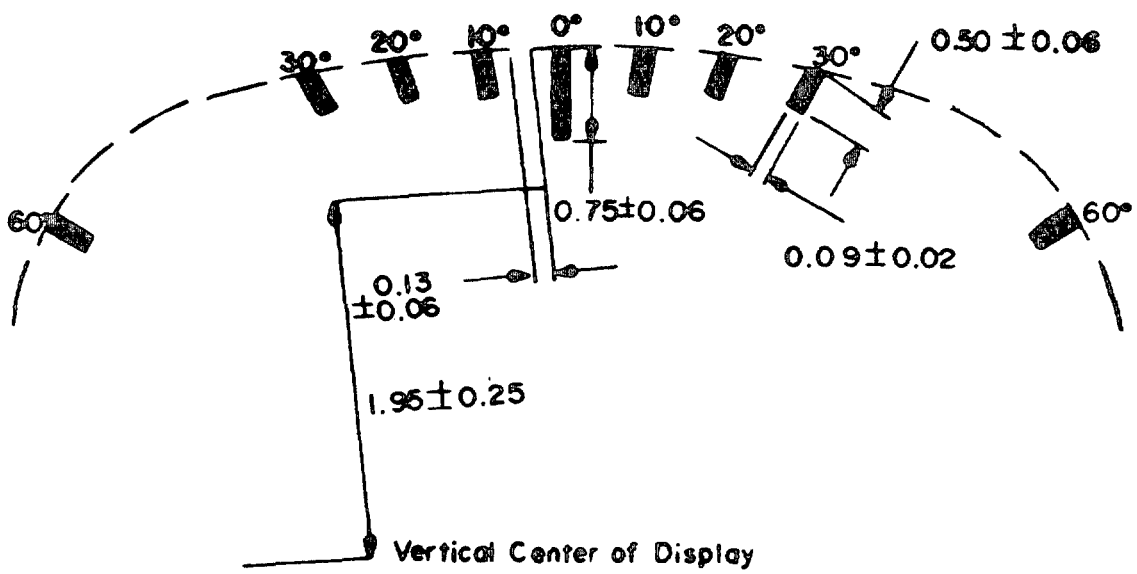
3.4.21 CHL generator. The CHL generator shall generate a presentation such that three heading lines appear within the flight path. Command heading lines shall be displayed during the flight mode and shall appear when a discrete +28 VDC signal is sent to the CHL generator. When in the landing or takeoff modes, this signal shall go to zero VDC (open circuit) resulting in the removal of the heading lines and the presentation of a path centerline as specified in 3.4.20.1. (See table VII, input 112.)

3.4.21.1 Vanishing point. The vanishing point of the flight path and command heading lines shall be displaced vertically and horizontally with respect to center of the effective display area. The vanishing point vertical and horizontal displacement signals shall orient both the path and command heading presentations.

3.4.21.2 CHL. A minimum of three black command lines shall appear on the illuminated path area. These lines shall be 0.06 ± 0.03 inches wide and shall appear to converge in an area just above the apex of the flight path. The lines shall be spaced such that any two lines shall enclose an area equal to one-third that of the flight path.

3.4.21.2.1 CHL horizontal motion. The analog dc input signal specified in 3.4.13.1.1.6 shall cause the CHL to move horizontally. An input signal of +15 VDC shall cause the lines to move at a maximum speed to the right. An input signal of -15 VDC shall cause the lines to move at a maximum speed to the left. The maximum speed to the right or left shall not exceed 2.45 inches per second. The voltage required to set the lines in motion in either direction shall be not greater than 0.7 VDC.

3.4.22 Roll pointer. A roll pointer shall be provided which will enable the pilot to read roll angles directly from the ADI. A video generated roll pointer shall roll with the display raster and shall be referenced to the reference markers described in 3.4.22.3. (See figure 20.)



Note: The reference marker dimensions shown are with the day/night filters are not installed.

FIGURE 20. Roll pointer and reference markers.

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3.4.22.1 Display. The roll pointer shall be visible in CA, TC, and test modes only and shall be accurate to ±2° roll as measured from the center of the pointer to the center of the reference markers.

3.4.22.2 Size. A bright line 0.13 ± 0.06 inches in width shall extend down from the top of the screen and its lower edge will be 1.95 ± 0.25 inches from the display center.

3.4.22.3 Reference markers. The reference markers shall have a dull black surface and shall be located as to minimize parallax. The markers shall be placed along the top edge of the CRT pointing toward the center of the display and spaced angularly from the top horizontal center at 0°, +10°, +20°, +30°, +60°, -10°, -20°, -30° and -60°.

3.4.22.3.1 Reference marker size. All markers shall be 0.09 ± 0.02 inches wide. The zero degree marker shall be 0.75 ± 0.06 inches long and the others shall be 0.50 ± .06 inches measured at the center of the lines.

3.4.23 Pitch trim reference cursors. Two movable pitch trim cursors extending from the sides of the display shall be provided to enable the pilot to maintain a constant elevation attitude, when required attitude is other than horizontal. The cursors shall roll with the horizon. (See figure 21.)

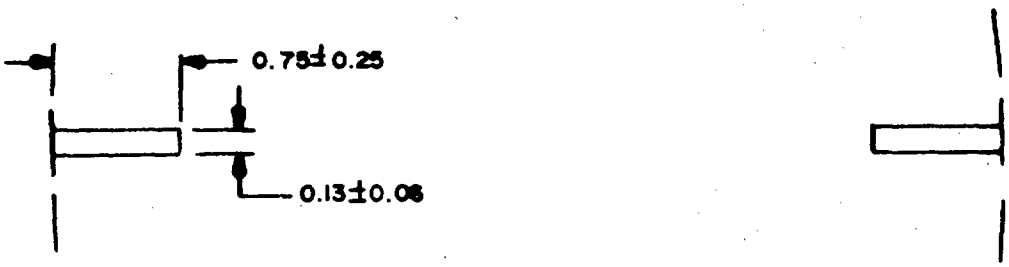


FIGURE 21. Pitch trim reference cursors.

3.4.23.1 Display. The pitch trim cursors shall be visible in CA and test modes, and in TC mode when input 33(b) is present. (See table I and 3.4.13.4.3.)

3.4.23.2 Adjustment. A potentiometer located on the pilot's control panel shall be used to control the vertical position of the pitch trim cursors with reference to the fiducial markers at zero degree roll. (See 3.5.1.4.7.4.)

3.4.23.2.1 Cursor size. The cursor lines at zero degree roll shall extend in from the display sides 0.75 ± 0.25 inches and shall be 0.13 ± 0.06 inches wide.

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3.4.23.2.2 Cursor accuracy. The cursor lines shall be parallel to the horizon within ±0.5° at zero degree roll. Zero VDC shall displace the pitch trim cursor 2.09 ± 0.29 inches above the horizontal centerline. A +10 VDC input shall place the cursors within ± 0.12 inches of the nearest fiducial marker. A +20 VDC input shall displace the cursors 2.09 ± 0.29 inches below the horizontal centerline and +30 Vdc shall displace the cursors below the display mask where they shall not be visible on the display.

3.4.24 Offset impact point marker. The offset impact point marker shall be a black rectangular horizontal bar with a bright square at the left and right ends. The marker shall be displayed in TC mode only. The offset impact point marker shall be used to assist the pilot in following the terrain while in TC mode. (See figure 22.)

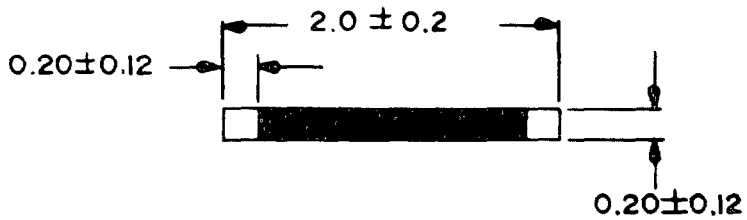


FIGURE 22. Offset impact point marker.

3.4.24.1 Size. The total offset impact point marker length shall be 2.0 ± 0.2 inches and the height shall be 0.20 ± 0.12 inches. The bright square at both ends shall be 0.20 ± 0.12 inches in length and height.

3.4.24.2 Scale factor. Vertical displacement of the impact point symbol shall result in the offset impact point marker moving either twice the vertical distance or the same vertical distance as the impact point.

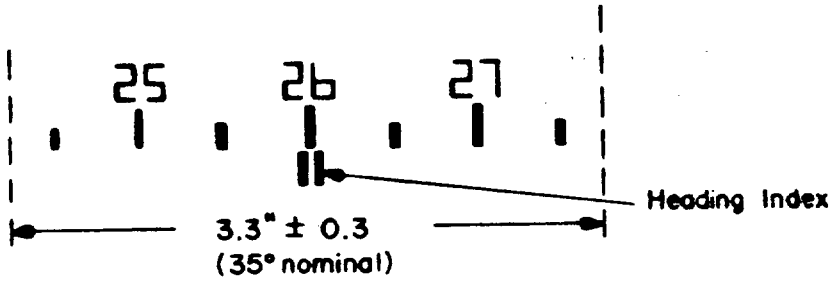
3.4.24.2.1 Twice vertical distance. The marker shall be capable of being manually positioned vertically by the pitch trim vertical positioning control and shall also move in response to the impact point vertical positioning voltage at twice the vertical rate of the impact point. This condition applies when input 33(b) is absent.

3.4.24.2.2 Same vertical distance. The marker shall be capable of being manually positioned vertically by the offset impact point marker vertical position control and shall also move in response to the impact point vertical positioning voltage at the same vertical rate as the impact point. This condition applies when input 33(b) is present.

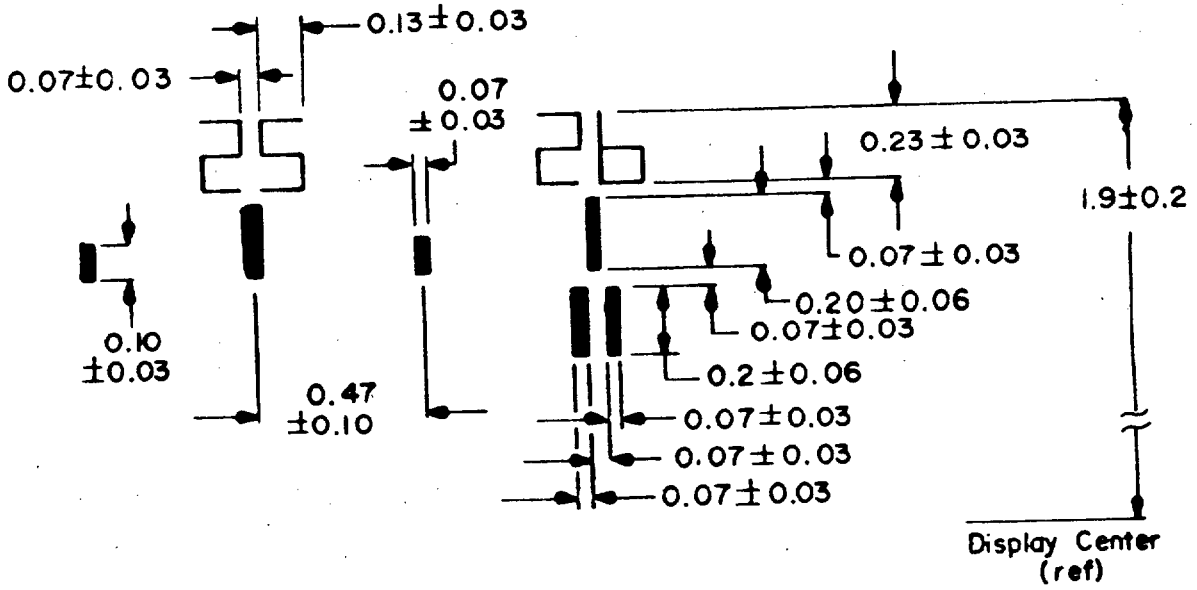
3.4.24.3 Position. With the impact point set at the display center and the pitch trim or offset impact point marker vertical positioning voltage adjusted between zero and +10 VDC inclusive, the offset impact point marker position shall be 0.87 ± 0.12 inches below the display center. When the pitch trim or offset impact point marker vertical positioning voltage is readjusted for +20 VDC, the offset impact point marker shall be 1.74 ± 0.25 inches below the display center.

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(A) Scale



(B) Scale tolerances



Notes:
1. All numerics shall be 2-digit groups including most significant zeroes where applicable.
2. All numerics shall have a nominal line width of 0.03 inches.

FIGURE 23. True/magnetic heading scale.

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3.4.25 True/magnetic heading scale. A scale showing aircraft true or magnetic heading shall be provided as shown in figure 23. It shall consist of a moveable scale, associated numerals, and an indexer. The indexer shall serve as the index mark indicating the true or magnetic heading of the aircraft as obtained from inputs 97 through 100. Heading range shall be 360 degrees, presented in an approximate aspect ratio of 1.5:1 with respect to the 50 degree azimuth field of view of the ADI.

3.4.25.1 Heading accuracy. Static error shall be not greater than ± 1 degree with respect to the heading index. Dynamic error shall be not greater than two degrees for heading rates up to and including 10 degrees per second. This error may be exceeded for rates in excess of 10 degrees per second. The ADI heading scale shall be capable of following a continuous rate of 10 degrees per second.

3.4.25.2 Heading scale position. The top of the scale shall be located 1.9 ± 0.2 inches above the display center. The heading index shall be positioned to within ± 0.125 inches of the display horizontal center.

3.4.26 Angle of attack symbol. An AOA reference symbol and an error symbol shall be provided as shown in figure 24. It consists of a fixed reference and a sidewise "T" error symbol which moves vertically about the fixed reference as a function of input 101. The vertical deviation of the symbol represents the AOA error which is determined by the difference between an internally preset angle and input 101.

3.4.26.1 AOA accuracy. When the "T" is aligned with the middle bar of the reference symbol, the AOA shall be 22 ± 0.16 units AOA.

3.4.26.2 Error symbol position. The right end of the "T" symbol shall be 2.0 ± 0.29 inches to the left of display horizontal center. The middle horizontal bar of the reference symbol shall be within ± 0.125 inches of the display vertical center.

3.4.26.3 Error symbol range. The "T" symbol shall not be limited and shall move vertically with a scale factor of 0.2 inches per unit AOA. The damping time constant shall be one second.

3.4.27 Radar altitude scale. A linear scale showing aircraft radar altitude shall be provided as shown in figure 25. The symbol consists of a thermometer type vertical scale with fixed indices and numerics.

3.4.27.1 Altitude scale position. The bottom of the left edge of the lowermost tick mark shall be located 1.87 ± 0.15 inches and 2.1 ± 0.2 inches below and to the right of the display center, respectively.

3.4.27.2 Altitude thermometer range. The altitude scale thermometer shall move from 0 to 2.5 representing an altitude from 0 to 2,500 feet. When +28 VDC (input 5) from the RDC is present the thermometer and numerics shall be blanked. The scale shall also be blanked when inputs 83 and 84, represented by connector pin numbers P2-36 and P2-40, are connected together in the aircraft and input 5 is present.

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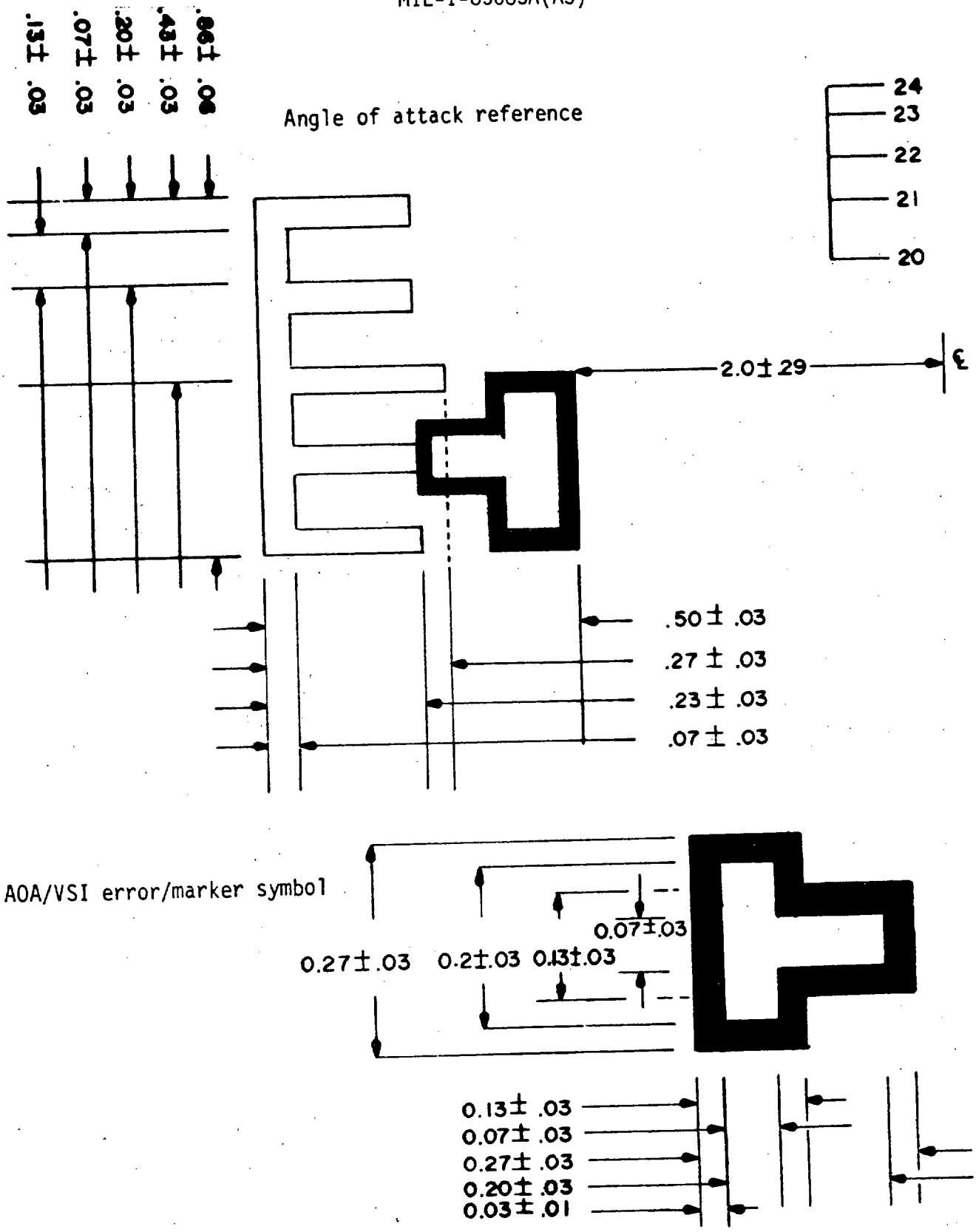
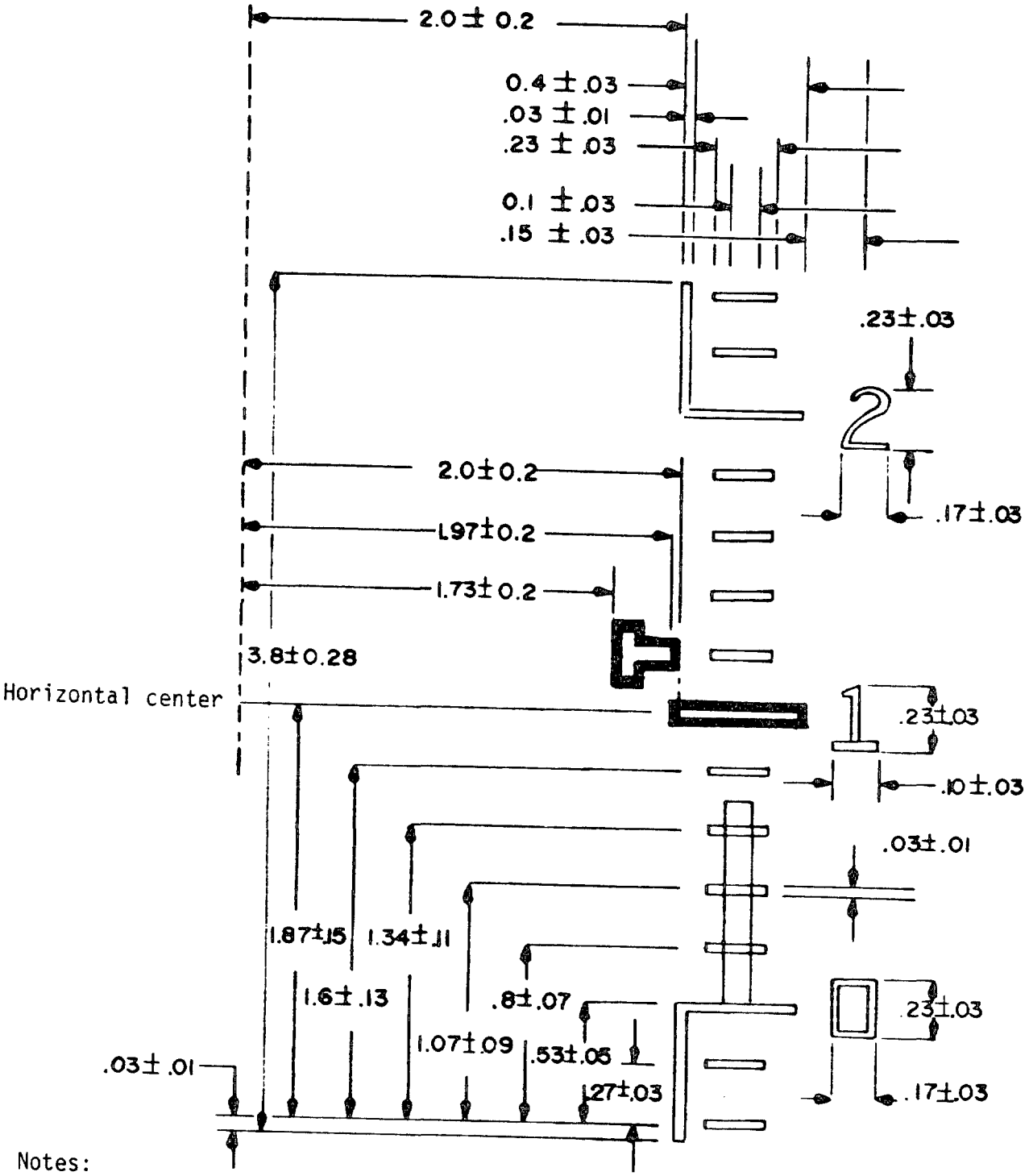


FIGURE 24. Angle of attack reference/error and VSI symbols.

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Notes:

1. Radar scale numerals shall be centered about their appropriate tick within +0.08 inches.
2. All numerics shall be equal in size with a nominal line width of 0.03 inches.

FIGURE 25. Altitude/VSI scale.

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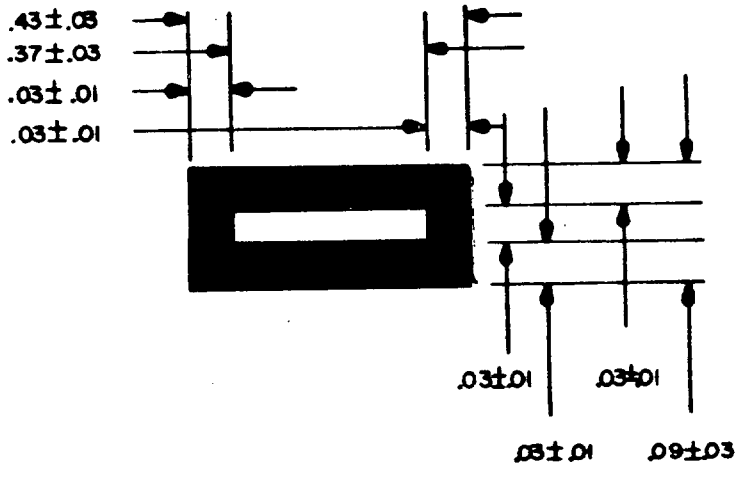


FIGURE 25. Altitude/VSI scale. - Continued

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3.4.28 Vertical speed indicator. A VSI scale and a marker shall be provided as shown in figures 24 and 25. The symbology consists of horizontal "T" marker which moves vertically about the combination VSI/radar altitude scale as a function of input 85. When inputs 83 and 84, represented by connector pin numbers P2-36 and P2-40, are connected together in the aircraft, the VSI marker shall not be displayed. The vertical position of the "T" marker with respect to the fixed scale tick marks shall represent the aircrafts vertical speed. The uppermost scale tick mark shall indicate +1400 feet per minute and the lowest scale tick mark shall indicate -1400 feet per minute. The zero reference for the VSI scale shall be the 1,000 foot tick (large 1) on the radar altitude scale which is located at the vertical center of both the VSI/radar altitude scale and the display. During the TC mode of operation the VSI/radar altitude scale shall be displayed.

3.4.28.1 VSI accuracy. When the "T" marker is aligned with the zero reference of the VSI scale, the vertical speed error shall be ± 25 feet per minute.

3.4.28.2 VSI marker position. The left end of the VSI marker shall be located 1.73 ± 0.2 inches to the right of the horizontal center.

3.4.28.3 VSI marker range. The VSI marker shall be limited to movement approximately 0.25 inches above and below the uppermost and lowest tick marks on the VSI scale. The VSI marker shall move vertically with a scale factor of 1.34 x 10^{-3} inches per feet per minute. The damping time constant shall be one second.

3.4.28.4 VSI marker vertical position. An input signal of ± 7.5 V shall be equivalent to a vertical speed of 1,000 feet per minute. A positive signal shall cause the marker to move down; a negative signal shall cause the marker to move up.

3.4.29 ACLS/ILS steering. The ADI shall provide a steering symbol as shown in figure 26. It shall consist of two independently controlled bars, one vertical and one horizontal. (See 3.4.13.8.5 through 3.4.13.8.7.)

3.4.29.1 Vertical bar. The vertical bar shall move horizontally as a function of the horizontal input voltage. For a zero VDC input, the vertical bar axis shall be within the reference square. The vertical bar shall be centered from top to bottom with respect to the horizontal display center to within ± 0.2 inches.

3.4.29.2 Horizontal bar. The horizontal bar shall move as a function of the vertical input voltage. For a zero VDC input, the horizontal bar axis shall be within the reference square. The horizontal bar shall be centered from right to left with respect to the vertical display center to within ± 0.2 inches.

3.4.29.3 Reference square. The reference square shall be centered to within ± 0.25 inches of the display center and have the dimensions shown in figure 26.

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Steering symbol and reference square.

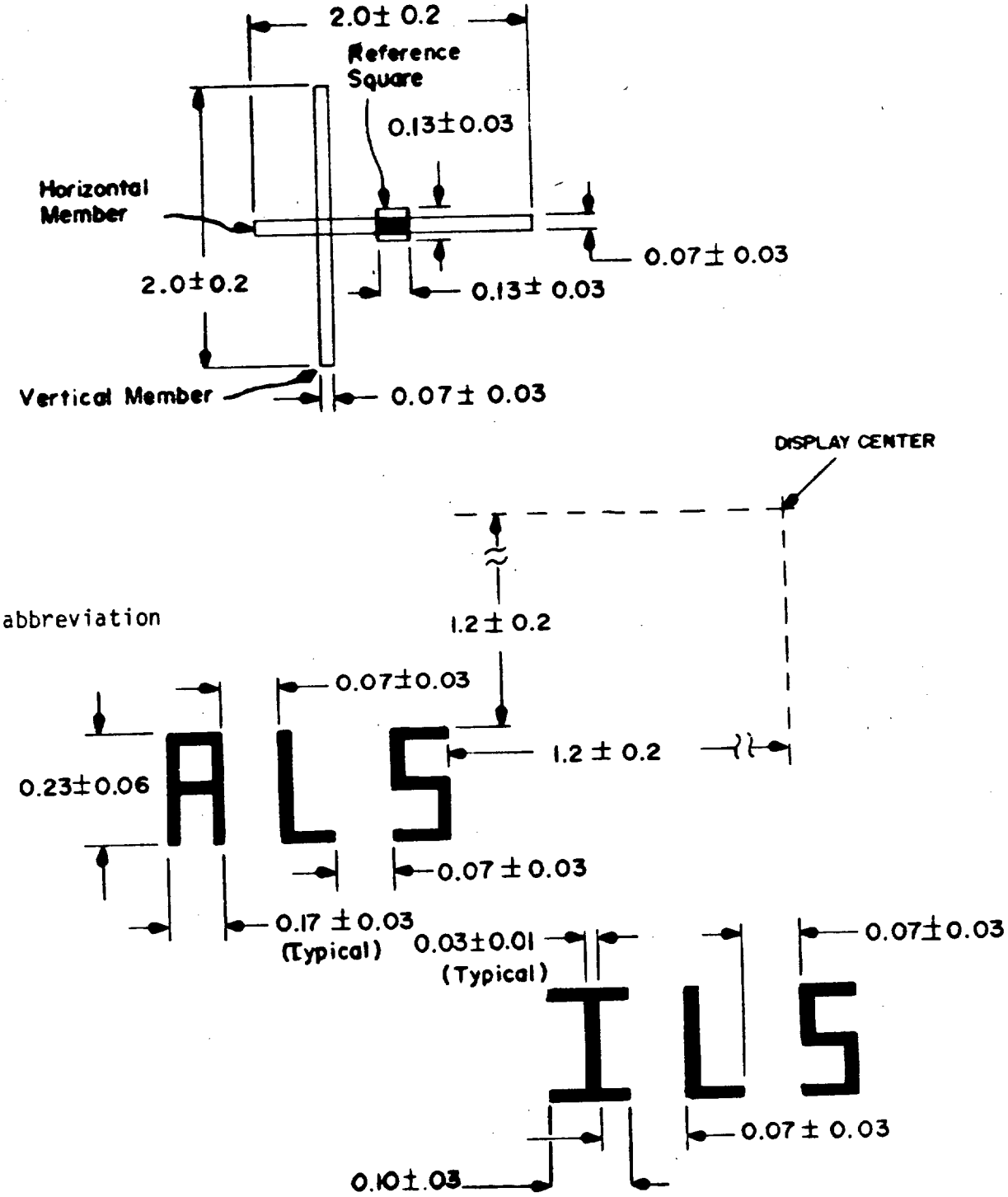


FIGURE 26. ACLS/ILS steering symbol.

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3.4.29.4 Brightness level. The vertical and horizontal bars and reference square shall be bright except for that portion of the reference square which is overlain by the horizontal or vertical bar which shall be black.

3.4.29.5 Mode abbreviation. The letters "ALS" or "ILS" shall be displayed in the lower left hand corner of the display. (See 3.4.13.8.7 and figure 26.) The extreme top of the right hand side of the "S" shall be located 1.2 ± 0.2 inches and 1.2 ± 0.2 inches to the left and below, respectively, of the display center.

3.4.29.6 Vertical and horizontal bar motion. For horizontal input voltages between -4.4 VDC and +4.4 VDC, the vertical bar shall move linearly between -1.0 ± 0.17 inches and +1.0 ± 0.17 inches to the left and right of the reference square, respectively. Horizontal input voltage of absolute magnitude greater that 4.4 VDC shall cause the vertical bar to limit. The horizontal bar shall move linearly in the vertical direction with the same scale factor as the vertical bar.

3.4.29.7 Bar accuracy. Bar position shall be measured from the long center axis of each bar. Allowable position error shall be 0.125 ± 0.08D inches, where D is the distance of the bar from the display center.

3.4.30 Contrast ratio. Contrast ratio shall be defined as the apparent symbol brightness divided by the apparent raster brightness. For measurement purposes the contrast ratio shall be the ratio of the measured brightness of the least bright symbol to the measured brightness of the raster when the symbol is blanked (turned off). This ratio shall be not less than 1.4, for a symbol brighter than the raster. In addition, a ratio not less than 1.4 shall apply between the brightness of adjacent shades as stated in 3.3.22.

3.4.31 Analog display contrast. The elements, flight path, and symbols that make up the analog display shall be readily distinguishable from one another. Table V contains the shades (relative brightness) corresponding to the various elements that make up the analog display.

3.4.31.1 Superimposed symbols. The logic of precedence of superimposed symbols shall be defined by the need for continuity in the display. The superposition of two or more symbols may result in a variation of the specified shading of the individual symbols; however, each of the superimposed symbols shall be individually discernible at the time of superposition. Exceptions are black-on-black and bright-on-bright symbols.

3.4.32 Declutter switch. A declutter switch shall be provided to blank the true/magnetic heading scale, the radar altitude/VSI scale with thermometer and numerics and the VSI marker in TC and CA modes. The switch shall be located on the front of the ADI as shown in figure 1. The switch shall not be lighted. Pushing the switch once shall remove the above symbology. Pushing the switch a second time shall restore the symbology.

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TABLE V. Symbol shades.

Shades	Symbols
Black	Zenith marker (+90° pitch line) Pull-up marker Release marker +60° pitch line +30° pitch line +20° pitch line +10° pitch line Flight path centerline and border Borders of: Weapon symbol Target symbol Impact point marker AOA error marker VSI marker and zero reference Offset impact point marker (center) Command heading lines Ground textures Horizon line
Medium dark	Ground plane
Medium light	Sky plane
Bright	Weapon symbol Target symbol Impact point marker Offset impact point marker (left and right ends) Roll marker Pitch trim cursors -40°, -30°, -20°, -10° pitch lines True/magnetic heading scale AOA reference and error symbol Radar altitude/VSI scale ACLS/ILS steering ALS or ILS mode alphanumerics VSI marker Radar altitude thermometer and numerics
Controllable (from transparent to medium light)	Flight path

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3.5 Detail requirements.

3.5.1 Analog display indicator. The ADI shall meet the requirements specified in 3.5.1.1 through 3.5.1.5.

3.5.1.1 Function. The ADI shall contain all the circuits and the CRT that provide the video display. The display shall contain elements which shall have a relationship and perspective to each other and to the attitude of the aircraft, permitting simulation of a realistic contact presentation.

3.5.1.2 Form factor. The dimensions of the ADI, actual mounting dimensions, location of connectors, nameplates, and marking information shall be as shown in figure 1.

3.5.1.3 Weight. The weight of the ADI including light filters shall be not greater than 47 pounds (21.3 kg).

3.5.1.4 Contents of ADI. The ADI shall contain the following assemblies, subassemblies, and circuits:

a. Video generator boards
b. Video amplifier
c. Low voltage power supply
d. High voltage power supply
e. CRT roll servo-assembly
f. CRT deflection assembly
g. Display tube and shield
h. Fan

3.5.1.4.1 Video generator boards. The video generator boards shall contain the circuitry required to electronically generate and control the vertical and horizontal sweeps, symbols, and markers. The deflection circuits shall provide sync pulses in a form compatible to the RDC and shall have the following characteristics in TC and CA modes only. (See 3.4.14.1.)

a. The ADI shall provide a 525 line raster with 2:1 interlace.
b. The frame repetition rate shall be 30 Hz. The field repetition rate shall be 60 Hz.
c. The vertical resolution shall not lose more than 5 percent of the horizontal lines during vertical retrace.
d. The linearity of the sweep shall be such as to produce a linear display within ± 5 percent.

3.5.1.4.2 Video amplifier. The video amplifier shall have a bandwidth of at least 10 MHz. The amplifier shall include an automatic gain control (AGC) to maintain the display control levels regardless of the FLIR video gain control setting or when switching between modes.

3.5.1.4.3 Low voltage power supply. The low voltage power supply shall function as an ac-to-dc convertor. It shall convert aircraft power to the power levels required for driving the ADI circuitry.

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3.5.1.4.4 High voltage power supply. The high voltage power supply shall supply the voltages required to operate the CRT.

3.5.1.4.5 CRT roll servo assembly. The roll servo assembly shall contain the yoke coils and a servo assembly capable of rotating the yoke assembly to roll the raster per roll input information. When in the FLIR or external mode of operation, the roll servo shall be electrically locked.

3.5.1.4.6 CRT deflection assembly. The deflection assembly shall contain circuitry to form current wave forms required to form the raster on the CRT.

3.5.1.4.6.1 Sweep direction. The direction of sweep as seen by an observer viewing the display face at a zero degree roll input for each mode shall be as specified in table VI.

TABLE VI. Sweep direction.

Mode	Field sweep	Line sweep
CA	Bottom to top	Left to right
TC	Bottom to top	Left to right
TRAM-FLIR	Left to right	Top to bottom
TRAM-external	Top to bottom	Left to right

3.5.1.4.7 Display tube.

3.5.1.4.7.1 CRT size. The CRT size shall be such that the minimum display size is not less than as follows:

a. Diagonal 7.66 inches
b. Height 5.23 inches
c. Width 7.04 inches

3.5.1.4.7.2 Persistence. The tube persistence shall be such that neither stickiness nor flicker of the display shall be perceptible to the human eye directly or under peripheral vision at conditions of bright daylight and night ambient illumination.

3.5.1.4.7.3 CRT phosphor. The CRT shall utilize a P43 phosphor.

3.5.1.4.7.3.1 Phosphor protection. Phosphor protection circuitry shall be provided such that loss of deflection shall not damage the phosphor.

3.5.1.4.7.4 Display reference markings. Two black fiducial markers shall be provided. One marker shall be located at each side of the display in such a manner as to minimize problems of parallax. The markers shall be in the form of an isosceles triangle with a 0.25 inches base and 0.25 inches height. The bases of the triangles shall be normal to the display horizon line at a zero degree roll angle. The bases of the triangles shall coincide with the left and right edges of the visible display area. A line drawn between the apexes of the triangles shall be parallel to the top and bottom edges of the ADI front panel and pass through the display center.

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3.5.1.4.7.5 Implosion shield. A transparent implosion shield shall be provided which shall be of sufficient strength to prevent particles from hitting the viewer, if implosion should occur. The shield shall be of such optical quality so as not to distort the image when viewed 25 inches away.

3.5.1.4.7.6 CRT general requirements. The CRT shall meet the requirements of MIL-E-1.

3.5.1.4.8 Fan. A 400 Hz fan may be incorporated to provide adequate cooling to permit proper ADI operation within the temperature range specified in 3.3.11.

3.5.1.5 Controls. An ADI shall not have any external controls necessary for operation except for the declutter switch. (See 3.4.32.) Controls related to modes of operation, test patterns, display brightness, contrast, path brightness, offset impact point position and pitch trim cursor display position shall be located on the pilot's control panel.

3.5.2 ADI filters. A day filter and a night filter shall be provided as part of the ADI. (See figure 1.) The filters shall be removeable and shall meet all the environmental conditions specified herein. One filter shall be stowed when the other is in use.

3.5.2.1 Day filter. The day filter shall have a one-quarter anti-reflective coating. (See 3.3.18.) There shall be no loss in resolution when viewing the CRT at a distance of 25 inches. The viewing cone shall be not less than of 30 degrees and shall provide cross viewing.

3.5.2.2 Night filter. The night filter shall be aviation red in color in accordance with MIL-C-25050. The loss of resolution shall be undectable by the human eye when viewing the CRT at a distance of 25 inches.

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TABLE VII. Input channel characteristics.

Input channel number	Received from	Voltage range	Type	Function	Input impedance Zin (Min)ohms	Connector and pin
1	Search radar	0/28	dc	Search radar TC warning lamp	Five (5) lamps in parallel MS 3338-6839; 28 V, 0.024A; 1166 (ea. lamp)	P1-49
2	Ballistic Computer	0/28	dc	0 Vdc removes CHL and displays path centerline	20K	P1-35
3	-	-	-	Radar signal ground	Floating return for RDC video (not used in A/C)	(Coax) P2-12
4	RDC	2 V_{pp}	Video wave-form	Video	100	(Coax) P2-48
5	RDC	0/28	dc	Radar altitude (RA) warning lamp	Five (5) lamps in parallel (see input 1)	P1-58
6	-	0 to 5	ac	(Not used)	-	P1-11
7	Ballistics computer	0/28	dc	ATTACK lamp	Three (3) lamps in parallel (see input 1)	P1-92
8	Ballistics computer	0/28	dc	RNG/LASER lamp	Three (3) lamps in parallel (see input 1)	P1-93
9	Ballistics computer	0/28	dc	Breakaway lamps	Eight (8) lamps in parallel (four each indicator) (see input 1)	P1-97
10	-	-	-	Legend lamps ground: (TC. RA, ATTACK, RNG/LASER breakaway, Computer	Floating return (to dimmer)	P1-98

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TABLE VII. Input channel characterstics. -Continued

Input channel number	Received from	Voltage range	Type	Function	Input impedance Zin (min)-ohms	Connector and pin
11	BAGS	-15 to +15	dc	BAGS weapon symbol horizontal position	10K	P1-74
12	BAGS	-15 to +15	dc	BAGS weapon symbol vertical position	10K	P1-75
13	BAGS	-	-	Gnd	Internal tie point connected to chassis	P1-73
14	BAGS	0/28	dc	BAGS weapon symbol off-on	2K	P1-29
15	Pilot's control panel	115-400 Hz Ph. 1	ac	Standby power	N/A	P1-1
16	Pilot's control panel	+28	dc	Standby power	(Not used in ADI)	P1-3
17	Pilot's control panel	115-400 Hz Ph. 1	ac	Primary power	N/A	P1-5
18	Pilot's control panel	115-400 Hz Ph. 2	ac	Primary power	N/A	P1-6
19	Pilot's control panel	115-400 Hz Ph. 3	ac	Primary power	N/A	P1-7
20	Pilot's control panel	+28	dc	Primary power	N/A	P1-8
21	Pilot's control panel	0/28	dc	Video group 1 off-on	2K	P1-16, 9, 12

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TABLE VII. Input channel characterstics. -Continued

Input channel number	Received from	Voltage range	Type	Function	Input impedance Zin (min)-ohms	Connector and pin
22	Pilot's control panel	0/28	dc	Video group 2 off-on	2K	P1-10, 17
23	Pilot's control panel	0/28	dc	Radar video off-on	2K	P1-13, 14
24	Not applicable	0/28	dc	Roll reference off-on	1K	P1-15
25	Pilot's control panel	0/30	dc	Test position off-on	1K	P1-18
26	Pilot's control panel	0/-30	dc	Test position off-on	1K	P1-19
27	Pilot's control panel	0 to 16	dc	Path intensity control wiper	10K	P1-22
28	Pilot's control panel	-	-	Path intensity control return	Gnd	P1-23
29	Pilot's control panel	0 to 15	dc	Raster intensity control wiper	Video amplifier op-amp 100K	P1-24
30	Pilot's control panel	-	-	Control panel Gnd	Internal tie point connected to chassis	P1-26
31	Pilot's control panel	0 to 30	dc	Contrast control	75K	P1-27
32	Pilot's control panel	-	-	Contrast control Gnd	Internal tie point connected to chassis	P1-30
33 (a)	Pilot's control panel	0 to 30	dc	Pitch trim cursor vertical position	100K	P1-32

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TABLE VII. Input channel characterstics. -Continued

Input channel number	Received from	Voltage range	Type	Function	Input impedance Zin (min)-ohms	Connector and pin
33 (b)	Pilot's control panel	0 to 30	dc	Offset impact point marker vertical position	100K	P1-59
34	INS	-	-	Pitch potentiometer Gnd	Internal tie point connected to chassis	P2-4
35	INS	-30 to +30	dc	Pitch horizon and DC signal	75K	P2-5
36	INS	-	-	Roll control transmitter neutral	Internal tie point connected to chassis	P2-23
37	INS	0 to 11.8	ac	Roll control transmitter- X	Stator-Kearfott P/N R1001-1A	P2-24
38	INS	0 to 11.8	ac	Roll control transmitter- Z	Stator-Kearfott P/N R1001-1A	P2-25
39	INS	0 to 11.8	ac	Roll control transmitter- Y	Stator-Kearfott P/N R1001-1A	P2-26
40	INS	0.3 to 0.8	ac	31.5 KC-variable phase	10K	P2-35
41	INS	-	-	Resolver Gnd	Internal tie point connected to chassis	P2-37
42	INS	-	-	Signal Gnd	Internal tie point connected to chassis	P2-52
43	INS	0/28	dc	PLATFORM lamp	Three (3) lamps in parallel (see input 1)	P2-50
44	INS	0/28	dc	+28 V filtered (roll)	2K	P2-51
45	INS	0/28	dc	+28 V unfiltered (pitch)	2K	P2-56

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TABLE VII. Input channel characterstics. -Continued

Input channel number	Received from	Voltage range	Type	Function	Input impedance Zin (min)-ohms	Connector and pin
46	INS	-	-	Gnd return for filtered and unfiltered +28 V	Floating return for fail safe relays	P2-55
47	Ballistics computer	0/28	dc	COMPUTER lamp	Three (3) lamps in parallel (see input 1)	P1-25
48	Ballistics computer	-15 to +15	dc	Path vanishing point horizontal position	8K	P1-36
49	Ballistics computer	-15 to +15	dc	Path vanishing point vertical position	10K	P1-37
50	Ballistics computer	0/28	dc	Path on-off	8K	P1-38
51	Ballistics computer	-15 to +15	dc	CHL rate control	14K	P1-39
52	Ballistics computer	-15 to +15	dc	Command horizontal position (path)	9K	P1-40
53	Ballistics computer	0 to +15	dc	Command altitude change	8K	P1-41
54	Ballastics computer	0/28	dc	Pull-up marker on-off	10K	P1-42
55	Ballistics computer	-15 to +15	dc	Pull-up marker position	12K	P1-43
56	Ballistics computer	0 to +15	dc	Path pull-up radius of curvature control	10K	P1-44
57	Ballistics computer	0/28	dc	Release marker on-off	7K	P1-45

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TABLE VII. Input channel characterstics. -Continued

Input channel number	Received from	Voltage range	Type	Function	Input impedance Zin (min)-ohms	Connector and pin
58	Ballistics computer	-15 to +15	dc	Release marker position	8K	P1-46
59	Ballistics computer	0 to +15	dc	Release marker size	10K	P1-47
60	Ballistics computer	0/28	dc	Target symbol on-off	25K	P1-50
61	Ballistics computer	-	-	Computer spare	--	P1-48
62	Ballistics computer	-15 to +15	dc	Target symbol vertical position	12K	P1-51
63	Ballistics computer	-15 to +15	dc	Target symbol horizontal position	10K	P1-52
64	Ballistics computer	-15 to +15	dc	Impact point vertical position	12K	P1-53
65	Ballistics computer	-15 to +15	dc	Impact point horizontal position	10K	P1-54
66	Ballistics computer	0/28	dc	Weapon symbol on-off	10K	P1-60
67	Ballistics computer	-15 to +15	dc	Weapon symbol horizontal position	10K	P1-61
68	Ballistics computer	-15 to +15	dc	Weapon symbol vertical position	12K	P1-62
69	Ballistics computer	0/28	dc	Stall signal off-on	10K	P1-63

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TABLE VII. Input channel characterstics. -Continued

Input channel number	Received from	Voltage range	Type	Function	Input impedance Zin (min)-ohms	Connector and pin
70	Ballistics computer	-	-	Computer Gnd	Internal tie point connected to chassis	P1-64
71	Ballistics computer	-	-	Computer Gnd	Internal tie point connected to chassis	P1-65
72	Ballistics computer	-	-	Computer Gnd	Internal tie point connected to chassis	P1-66
73	Ballistics computer	0/28	dc	Path inversion off-on	3.3K	P1-67
74	N/A	0/28	dc	Pitch trim cursor on-off	6.8K	P1-2
75	-	-	-	Aircraft Gnd	Internal tie point connected to chassis	P1-4
76	-	-	-	A/C wiring shields tie point	Internal tie point connected to chassis	P1-88
77	-	-	-	Spare	-	P1-55
79	-	-	-	Spare	-	P1-76
80	-	-	-	Spare	-	P1-79
81	-	-	-	Spare	-	P1-82
82	-	-	-	Spare	-	-
83	Aircraft	-	-	Jumper to 84	10	P2-36
84	Aircraft	-	-	Jumper to 83	10	P2-40
85	Ballistics computer	-15 To +15	dc	VSI position	20K	P2-21

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TABLE VII. Input channel characterstics. -Continued

Input channel number	Received from	Voltage range	Type	Function	Input impedance Zin (min)-ohms	Connector and pin
86	Ballistics computer	-	-	VSI return	Gnd	P2-19
87	-	-	-	Spare	-	P2-54
88	-	-	-	Dead end wires	-	P1-87
89	-	-	-	Dead end wires	-	P1-90
90	-	-	-	Dead end wires	-	P1-91
91	-	3.0 V_{pp}	Video	External video	93	P2-41
92	-	-	-	External video return	Gnd	P2-42
93	DRS	3.70	Pulse	FLIR vertical sync	130	P2-45
94	DRS	3.70	Pulse	FLIR horizontal sync	130	P2-49
95	DRS	0/28	dc	TRAM off-on	25K	P2-38
96	-	0/28	dc	External video off-on	25K	P2-39
97	Synchro transmitter	0-11.8	ac	Heading transmitter X	Synchro stator	P2-29
98	Synchro transmitter	0-11.8	ac	Heading transmitter Z	Synchro stator	P2-27
99	Synchro transmitter	0-11.8	ac	Heading transmitter Y	Synchro stator	P2-28
100	AOA probe	-	-	AOA return	Gnd	P2-30
101	AOA probe	0-10.0	dc	AOA	100K	P2-1

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TABLE VII. Input channel characterstics. -Continued

Input channel number	Received from	Voltage range	Type	Function	Input impedance Zin (min)-ohms	Connector and pin
102	Radio Altimeter	+1.0 to +25.0	dc	Radar altitude	100K	P2-2
103	ACLS or ILS	-20.0 to +20.0	dc	Vertical cross pointer deviation	100K	P2-8
104	ACLS or ILS	-20.0 to +20.0	dc	Horizontal cross pointer deviation	100K	P2-14
105	Aircraft	0/28	dc	Cross pointers validity	10K	P2-15
106	Aircraft	-	-	Return for inputs 103 thru 105	Gnd	P2-16
107	DRS	12.3 V_{pp}	Video	FLIR video	75	P2-13
108	DRS	0 to 1.4	Pulse	FLIR interlace	75	P2-31
109	DRS	-	-	FLIR interlace return	Gnd	P2-32
111	Radar altimeter	-	-	Radar altimeter return	Gnd	P2-33
112	Aircraft	0/28	dc	Landing mode off-on	2K	P2-18
113	Aircraft	0/28	dc	ACLS/ILS mode select	2K	P2-20
114	Aircraft	-	-	Jumper to 115	10	P2-43
115	Aircraft	-	-	Jumper to 114	10	P2-44

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TABLE VIII. Output channel characteristics.

Output channel number	Routed to	Voltage range	Type	Function	Load impedance ZL	Connector and pin
1	Pilot's control panel	-30	dc	B minus	1K	P1-20, 78
2	Pilot's control panel	+30	dc	B plus	750	P1-21, 84
3	Pilot's control panel	+15	dc	Brightness potentiometer	100K	P1-31
4	RDC	5 V_{pp}	Pulse 60 Hz	Radar vertical sync	RG 195/U:test set 100Ω RDC 330Ω	Coax P2-46
5	RDC	5 V_{pp}	Pulse 15,750 Hz	Radar horizontal sync	RG 195/U:test set 100Ω RDC 220Ω	Coax P2-47
6	INS	+30	dc	Excitation for pitch potentiometer	6.25K	P2-6
7	INS	-30	dc	Excitation for pitch potentiometer	6.25K	P2-7, 3
8	INS	5 V_{pp}	ac	31,500 Hz fixed phase	Resolver: Kearfott 3R982-0002-0 or equivalent	P2-17
9	—	115 400 Hz Phase 1	ac	Test Point	Rotor Kearfott RS 911-6A	P2-22
10	—	+28	dc	Test Point	N/A	P2-53

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified herein. Except as modified herein, inspections to demonstrate compliance with requirements of MIL-E-5400 shall use the procedures of MIL-T-5422.

4.4 First article inspections. First article inspection shall be conducted by the contractor on two ADIs representative of the production ADIs to be supplied under the contract. The Government inspector and the procuring activity shall be advised 10 days in advance of when inspections are to be conducted so that a Government representative may be designated to witness or supervise the inspections when so desired.

4.4.1 First article inspection procedures. First article inspection procedures shall be prepared as specified in 4.6.

4.4.2 First article inspection report. The contractor shall prepare a first article inspection report containing all the data collected in conducting these inspections (see 6.2).

4.4.3 Scope of tests. First article tests shall consist of the following tests:

a. Bench tests (see 4.4.3.1)
b. Environmental (see 4.4.3.2)
c. Electromagnetic interference (see 4.4.3.3)
d. Maintainability demonstration (see 4.4.3.4)
e. Reliability qualification (see 4.4.3.5)

4.4.3.1 Bench tests. The individual tests specified in 4.5.11 shall be conducted at the standard conditions specified in 3.3.9.

4.4.3.2 Environmental. Except as modified herein, the environmental tests shall be performed in accordance with MIL-T-5422 for class 1 equipment. Prior to, during and after each environmental test, the requirements of this specification shall be verified. (See table IX.)

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TABLE IX. Environmental performance verification tests.

Test	Requirement Paragraph
Warm-up	(3.3.10)
Switching time	(3.4.3)
Inertial subsystem reliability indicator	(3.4.6)
Computer reliability indicator	(3.4.7)
Search radar TC reliability indicator	(3.4.8)
Altimeter reliability indicator	(3.4.9)
RNG/LASER indicator	(3.5.10)
Attack Indicator	(3.4.11)
Breakaway Indicator	(3.4.12)
FLIR	(3.4.13.7.1)
External video	(3.4.13.7.2)
Outputs	(4.4.14)
Display brightness	(3.4.15.2)
Display resolution and contrast	(3.4.15.3)
Sky ground plane	(3.4.15.3.3.1)
Ground texture generator	(3.4.16)
Symbol generators	(3.4.17)
Pitch lines	(3.4.18)
Horizon line	(3.4.19)
Flight path generator	(3.4.20)
CHL generator	(3.4.21)
Roll pointer	(3.4.22)
Pitch trim reference cursors	(3.4.23)
Offset impapct point marker	(3.4.24)
True/magnetic heading scale	(3.4.25)
AOA symbol	(3.4.26)
Radar altitude scale	(3.4.27)
Vertical speed indicator	(3.4.28)
ACLS/ILS steering	(3.4.29)
Declutter switch	(3.4.32)

4.4.3.2.1 Temperature and altitude. The ADI shall operate within required limits when tested in accordance with MIL-T-5422, class 1, with upper altitude limit modified to 55,000 feet.

4.4.3.2.2 Thermal shock. The ADI shall operate within required limits when tested in accordance with MIL-T-5422.

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4.4.3.2.3 Cooling performance test. The cooling performance test for the ADI shall be conducted at the levels shown in table X. Thermal steady state shall be achieved during all cooling tests. The temperatures of all components, within the ADI, which are expected to operate near their limiting values shall be monitored and recorded during these tests. The ADI shall be electrically monitored to assure satisfactory operation during the cooling tests, under all modes of operation. The contractor may choose to substitute, subject to the procuring activity approval, other test levels for critical cooling conditions.

TABLE X. Cooling test level.

Test No.	Ambient Temp. (°F)	Ambient Pressure (psia)
1	131	14.67
2	127	10.9
3	106	5.5
4	68	2.25

4.4.3.2.4 Vibration. The ADI shall be tested in accordance with curve 1 of figure 2 of MIL-E-5400, modified to 5G.

4.4.3.2.5 Shock. The ADI shall be tested in accordance with MIL-T-5422, figure 10, using a half-sine wave.

4.4.3.2.6 Acceleration. The ADI shall be tested in accordance with procedure III of MIL-E-5272. Functional tests during this environmental test shall be made by monitoring the test mode on the ADI.

4.4.3.2.7 Humidity. The ADI shall be tested in accordance with MIL-T-5422.

4.4.3.2.8 Explosive decompression. The ADI shall be subjected to tests simulating cabin blow-out conditions resulting from the explosive decompression rates for conditions I and II as shown in figure 27. During these tests, the chamber and ADI internal pressure variations shall be recorded. At the conclusion of the condition 1 test, the ADI shall meet the requirements specified herein. At the conclusion of the condition II test, the ADI shall not be required to operate as specified.

4.4.3.2.9 Salt spray. The ADI shall be tested in accordance with MIL-T-5422.

4.4.3.2.10 Explosion. The ADI shall be tested in accordance with MIL-T-5422.

4.4.3.2.11 Sand and dust. The ADI shall be tested in accordance with MIL-T-5422.

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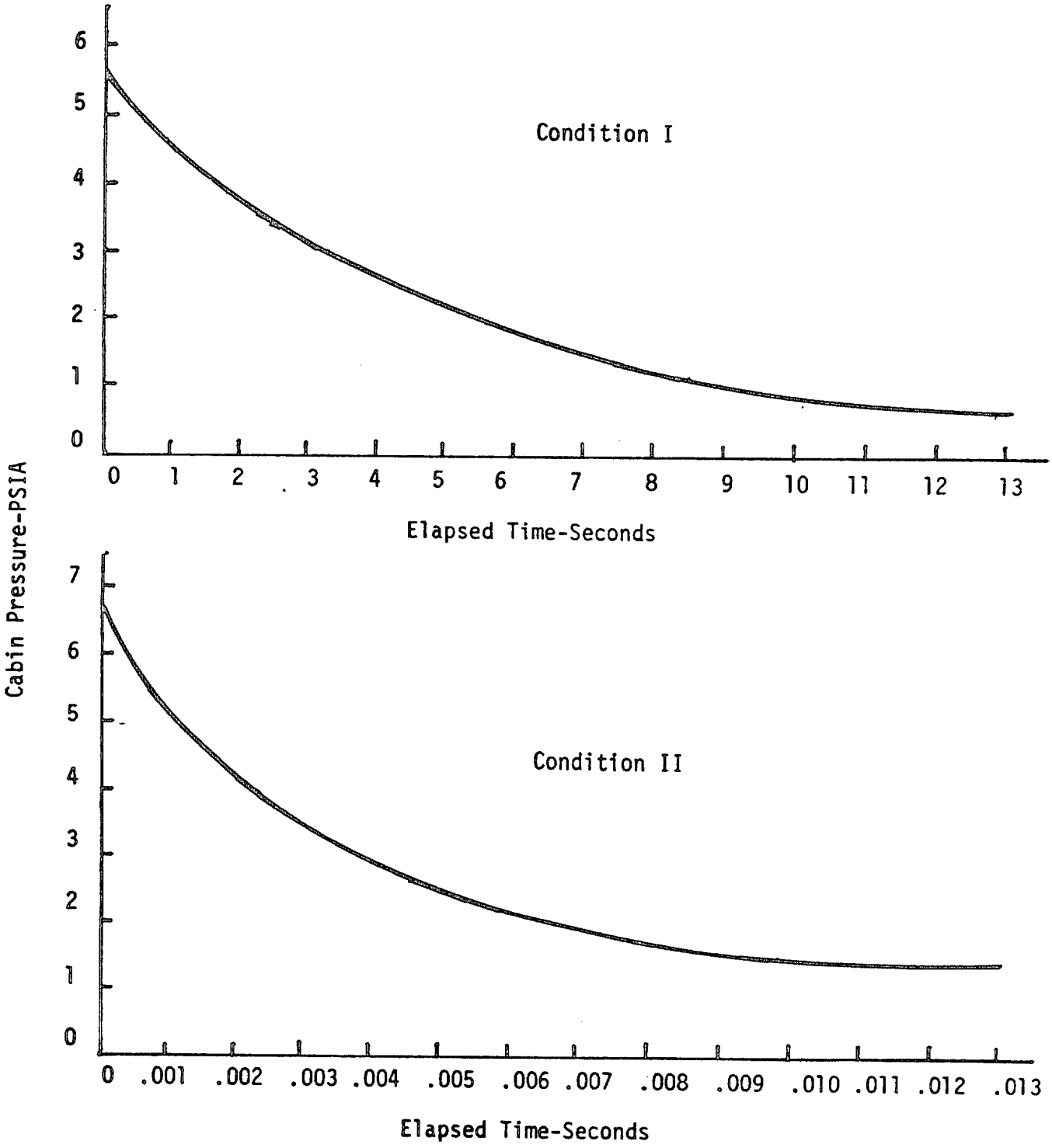


FIGURE 27. Cabin pressure decay during explosive decompression.

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4.4.3.2.12 Fungus. The ADI shall be tested in accordance with MIL-T-5422.

4.4.3.2.13 Acoustic noise. The ADI shall be tested in accordance with MIL-T-5422.

4.4.3.2.14 Power characteristics test. The ADI shall be tested to demonstrate compliance with the requirements of MIL-STD-704, with the additions specified in 4.4.3.2.14.1 and 4.4.3.2.14.2.

4.4.3.2.14.1 Power consumption test. The power required to operate the ADI shall be measured in the maximum power consumption mode and shall be not greater than the value specified in 3.3.12.1.

4.4.3.2.14.2 Under-voltage test. Reduce the line voltage from standard conditions to zero and then back to standard conditions over a period of five minutes. There shall be no damage to any components as a result of this test.

4.4.3.2.15 Environmental test combinations. When more than one unit is provided as the first article, tests shall be conducted on the units as specified in table XI.

TABLE XI. Environmental tests.

TEST	ADI Number
Temperature - altitude	1
Vibration	1
Shock	1
Thermal shock	1 or 2
Cooling performance	1 or 2
Acceleration	1 or 2
Explosive decompression	1 or 2
Sand and dust	1 or 2
Fungus	1 or 2
Acoustic	1 or 2
Power characteristics	1 or 2
EMI	2
Humidity	2
Salt fog	2
Explosion	2

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4.4.3.3 Electromagnetic interference test. The ADI shall be tested in accordance with MIL-STD-462.

4.4.3.4 Maintainability demonstration tests. A maintainability demonstration shall be conducted in accordance with MIL-STD-470 and MIL-STD-471. The test organization shall verify ADI compliance with the intermediate level maintenance requirements. The tests shall demonstrate that the BIT function and self test features are capable of fault detection and isolation to a WRA and its SRAs.

4.4.3.5 Reliability qualification test. Two ADIs shall be tested in accordance with MIL-STD-781. Test plan XIVC shall be used. The temperature shall be cycled from -54°C to +55°C as delineated in figure 28, except the exact time of each event shall be determined by thermal survey. Each ADI shall be normally mounted and subjected to random vibration as delineated in figure B-4 of MIL-STD-781, except W_0 shall be 0.0024. The duration of vibration shall be at least 30 minutes during each hour of "on time." Failures shall be classified in accordance with MIL-STD-2074.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspections. All inspections and testing shall be subject to monitoring by the Government inspector. The contractor shall prepare test reports showing quantitative results for all quality conformance inspections. Such reports shall be signed by an authorized representative of the contractor. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product.

4.5.1 Scope of tests. Quality conformance testing shall consist of the following tests:

a. Individual tests (see 4.5.1.1)

b. Manufacturing screening test (see 4.5.1.2)

c. Reliability acceptance test (see 4.5.1.3)

4.5.1.1 Individual tests. Each ADI submitted for acceptance shall be subjected to the individual tests. These tests shall demonstrate compliance with the requirement of material, workmanship, and operational adequacy. As a minimum, each ADI accepted shall have passed the following tests:

a. Examination of product

b. Operational tests.

4.5.1.1.1 Examination of product. Each ADI shall be examined carefully to determine that the material and workmanship requirements have been met, and the physical characteristics of the ADI agree with the outline dimensions specified herein.

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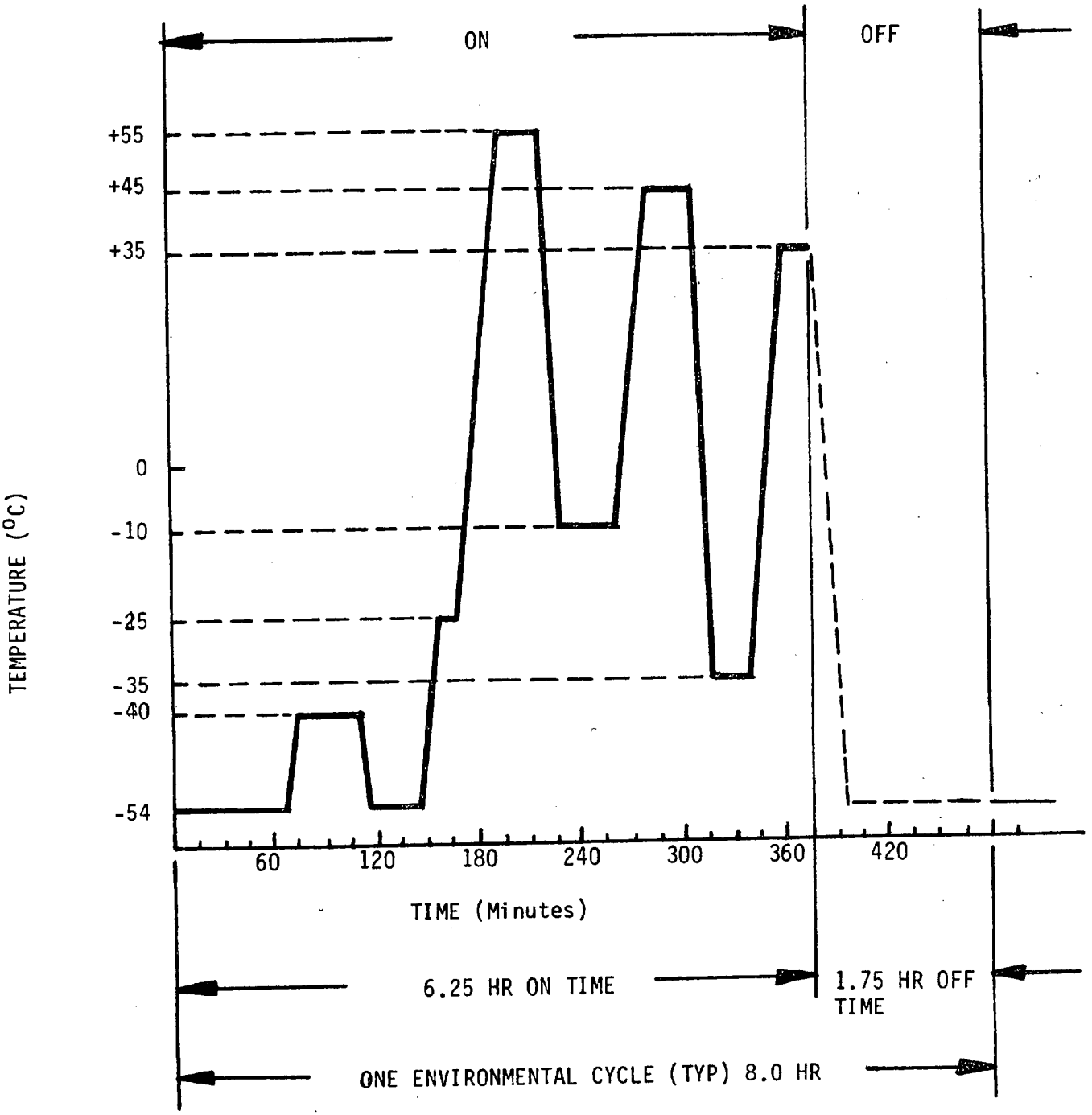


FIGURE 28. Reliability test environment.

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4.5.1.1.2 Operational tests. Prior to conducting the operational test each ADI shall be operated until the ADI temperature stabilizes. The requirements of this specification shall be verified. (See table XII.)

4.5.1.2 Manufacturing screening test. Each production ADI shall be subjected to random vibration in order to stimulate the early occurrence of failures due to manufacturing defects in parts and workmanship. The equipment shall be operational and monitored for proper performance during random vibration cycle.

4.5.1.2.1 Test duration. The duration of vibration shall be 10 minutes. The last five minutes shall be failure free.

4.5.1.2.2 Test axis. The ADI shall be hardmounted in the axis perpendicular to the printed circuit boards.

4.5.1.2.3 Test spectrum. The screening test shall consist of random vibration in accordance with Figure 29.

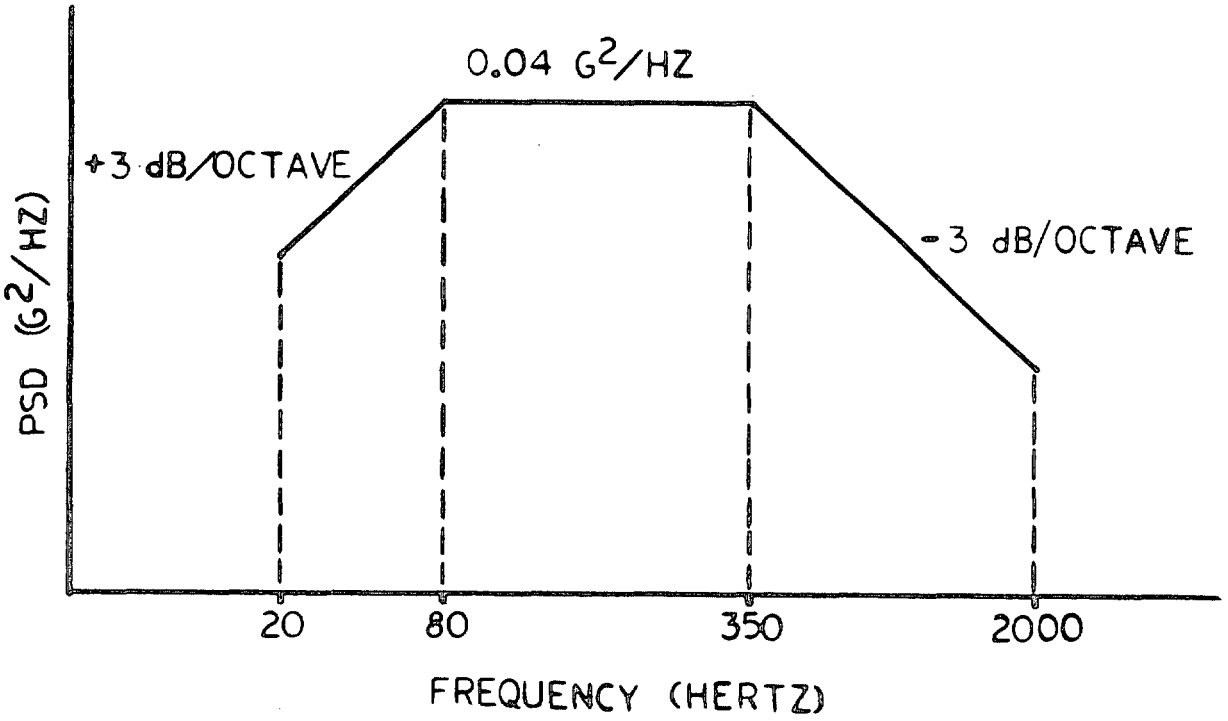


FIGURE 29. Random vibration.

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TABLE XII. Operational tests.

Test	Requirement paragraph
Warm-up time	3.3.10
Lighting power	3.3.12.2
Search radar TC display	3.4.2
Switching time	3.4.3
Inertial subsystem reliability indicator	3.4.6
Computer reliability indicator	3.4.7
Search radar TC reliability indicator	3.4.8
Altimeter reliability indicator	3.4.9
RNG/LASER indicator	3.4.10
Attack indicator	3.4.11
Breakaway indicator	3.4.12
FLIR	3.4.13.7.1
External video	3.4.13.7.2
Outputs	3.4.14
Display center	3.4.15.1
Display brightness	3.4.15.2
Display resolution and contrast	3.4.15.3
Sky plane	3.4.15.3.1
Ground texture generator	3.4.16
Symbol generators	3.4.17
Pitch line	3.4.18
Horizon line	3.4.19
Flight path generator	3.4.20
CHL generator	3.4.21
Roll pointer	3.4.22
Pitch trim reference cursors	3.4.23
Offset impact point marker	3.4.24
True/magnetic heading scale	3.4.25
AOA symbol	3.4.26
Radar altitude scale	3.4.27
Vertical speed indicator	3.4.28
ACLS/ILS steering	3.4.29
Declutter switch	3.4.32
Display reference markings	3.5.1.4.7.4

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4.5.1.3 Reliability Assurance Tests. Reliability Assurance Tests (RAT) shall be conducted in accordance with MIL-STD-781.

4.5.1.3.1 All equipment test. Each production ADI, except those submitted for the reliability qualification testing, shall be tested in accordance with the all equipment production reliability acceptance test of MIL-STD-781. Test plan XVIIIC shall be used with the accept-reject criteria derived from test plan IVC, shown in figure C-13 of MIL-STD-781. The test period for each production ADI shall be 50 failure free hours.

4.5.1.3.2 Test profile. RAT shall consist of repeated eight hour cycles as shown in figure 28. The vibration shall be in accordance with MIL-STD-781, except shall w_0 be 0.0024. The vibration shall be not less than 10 min/hr.

4.5.1.3.3 Failure classification. Failures shall be defined and classified in accordance with MIL-STD-781 and MIL-STD-2074.

4.5.1.3.4 Equipment burn-in. Prior to the 50 hour failure free test, the ADI shall be subjected to a burn-in period of not less than eight cycles using the test profile shown in figure 28.

4.5.1.3.5 Test MTBF. The upper test MTBF ($\emptyset_0$) shall be 445 hours and the lower test MTBF ($\emptyset_1$) shall be 223 hours.

4.5.1.3.6 Accept-reject criteria. Accept-reject criteria shall be in accordance with MIL-STD-781. At any time that the totals of test hours and test failures show a reject situation, the procuring activity shall be notified immediately. The procuring activity reserves the right to stop the acceptance of ADIs at any time that a reject situation exists.

4.5.1.3.7 Test details. The test details such as the length of test cycle, temperature and vibration profile, performance characteristics to be measured and any special failure criteria shall be part of the test procedure made available to the procuring activity for acceptance.

4.6 Test procedures. The procedures used for conducting first article tests and quality conformance tests shall be prepared by the contractor and made available to the procuring activity for review and acceptance (see 6.2.2). The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modifiication of such procedures, including additional test, when deemed necessary.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation

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for level A shall be determined in accordance with the selection chart in appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. The ADI is intended to display, to the pilot, all necessary flight cues to enable him to fly a complete mission from take-off to landing. The ADI will accomplish this by utilizing inputs from the peripheral sensor equipments (radar, air data computer, inertial subsystem, etc.) to present an integrated comprehensive display of flight information, using electronically generated symbology.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First Article, if required (see 3.1).

c. Selection of applicable levels of packaging and packing.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approval CDRL incorporated into the contract. When the provisions of FAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID
3.2.1	Nonstandard Part Approval Requests/Proposed Additions to an Approved PPSL	DI-E-7028A
3.3.2	Plan, Reliability Program	DI-R-7079
3.3.8	Requests, Nomenclature and Identification Assignment and Approval	UDI-E-21582 UDI-E-21584
4.6	Procedures, Test	UDI-T-21347
4.4.2	Report, First Article (Preproduction) Test	UDI-T-21349
4.4.3.4 4.4.3.5	Procedures, Maintainability and Reliability Demonstration Tests	DI-R-7035
4.4.3.4 4.4.3.5	Reports, Maintainability and Reliability Demonstration Tests	DI-R-7034
4.5.4	Plan, Corrective Action	DI-R-7038

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(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 First article. When a first article inspection is required, the item(s) will be tested and should be a first article sample. The first article should consist of two units. The Contracting Officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

6.4 Associated equipment. The ADI should operate and be compatible with the following associated equipments (which are part of the A6 aircraft) to the extent of their inputs and outputs as specified in tables VII and VIII. These associated equipments should not be supplied as part of the analog display system.

a.	Missile guidance and control system	AN/ARW-73
b.	Ballistics computer set	AN/ASQ-155, -61A, -133
c.	Pilot's control panel	128AV66804, 66826
d.	Radar set	AN/APQ-92, -148, -156
e.	Inertial navigation system	AN/ASN-31, -92
f.	Radar data converter	CV 1607/AVA-1
g.	Detecting and ranging set	AN/AAS-33A
h.	Radar altimeter	AN/APN-141, -194
i.	Angle of attack system	P/N TRU-109A
j.	Automatic carrier landing system/ instrument landing system	AN/ASW-25B, AN/ARA-63

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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6.6 Acronyms.

ACLS/ILS - Automatic Carrier Landing System/Instrument Landing System.
ADI - Analog Display Indicator.
AGC - Automatic Gain Control.
ALS - Short for ACLS.
AOA - Angle of Attack.
BAGS - Bullpup Automatic Guidance System.
BIT - Built-In-Test.
CA - Contact Analog.
CHL - Command Heading Lines.
CRT - Cathode Ray Tube.
DRS - Detecting and Ranging Set.
EMI - Electromagnetic Interference.
ERT - Equipment Repair Time.
FLIR - Forward Looking Infra Red.
GND - Ground.
GSE - Ground Support Equipment.
INS - Inertial Navigation System.
MTBF - Mean Time Between Failures.
PSI - Pounds per Square Inch.
PSIA - Pounds per Square Inch Absolute.
RA - Radar Altitude.
RDC - Radar Data Converter.
SACE - Shop Automatic Checkout Equipment.
SRA - Shop Replaceable Assembly.
TC - Terrain Clearance.
TRAM - Target Recognition Attack Multisensor.
VA - Volt Ampere
VAC - Volts Alternating Current.
VAST - Versatile Avionic Shop Test.
VDC - Volts Direct Current.
VGI - Vertical Gyro Indicator.
VSI - Vertical Speed Indicator.
WRA - Weapon Replaceable Assembly.

Preparing Activity
Navy-AS
(Project No. 5895-N 047)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-85083A(AS)	Military Specification Indicator, Analog Display

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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